



How Does Prior Distribution Affect Model Fit Indices of Bayesian Structural Equation Model?

Yonglin Feng¹ · Junhao Pan¹

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Abstract

Bayesian structural equation model (BSEM) integrates the advantages of the Bayesian methods into the framework of structural equation modeling and ensures the identification by assigning priors with small variances. Previous studies have shown that prior specifications in BSEM influence model parameter estimation, but the impact on model fit indices is yet unknown and requires more research. As a result, two simulation studies were carried out. Normal distribution priors were specified for factor loadings, while inverse Wishart distribution priors and separation strategy priors were applied for the variance–covariance matrix of latent factors. Conditions included five sample sizes and 24 prior distribution settings. Simulation Study 1 examined the model-fitting performance of BCFI, BTLI, and BRMSEA proposed by Garnier-Villarre and Jorgensen (*Psychol Method* 25(1):46–70, 2020) and the PPp value. Simulation Study 2 compared the performance of BCFI, BTLI, BRMSEA, and DIC in model selection between three data generation models and three fitting models. The findings demonstrated that prior settings would affect Bayesian model fit indices in evaluating model fitting and selecting models, especially in small sample sizes. Even under a large sample size, the highly improper factor loading priors resulted in poor performance of the Bayesian model fit indices. BCFI and BTLI were less likely to reject the correct model than BRMSEA and PPp value under different prior specifications. For model selection, different prior settings would affect DIC on selecting the wrong model, and BRMSEA preferred the parsimonious model. Our results indicate that the Bayesian approximate fit indices perform better when evaluating model fitting and choosing models under the BSEM framework.

Keywords Prior distribution · Bayesian structural equation model · Model fit indices · Bayesian approximate fit indices · Model selection

✉ Junhao Pan
panjunh@mail.sysu.edu.cn

¹ Department of Psychology, Sun Yat-Sen University, Guangzhou 51006, China

1 Introduction

In recent years, although the majority of research still employs frequentist methods in statistical analysis, an increasing number of researchers have begun to opt for Bayesian methods in their studies, and the advantages of Bayesian methods have gradually become more evident (Zondervan-Zwijenburg et al. 2017). Muthén and Asparouhov (2012) proposed the Bayesian structural equation model (BSEM) as an extension of Bayesian methods within the framework of the structural equation model (SEM). One notable feature of Bayesian methods is their utilization of priors, which incorporate information derived from previous research or expert advice, enabling valuable insight for parameter estimation.

However, when employing BSEM, an inevitable question arises about how to judge the accuracy of model fit and make selections among different models. The research on Bayesian model fit indices is still limited. Further investigation is needed to understand the sensitivity of different fit indices to the specification of prior distributions (Garnier-Villareal and Jorgensen 2020; Liu et al. 2022; Winter and Depaoli 2022a).

The present study investigated the prior specifications in the confirmatory factor analysis (CFA) within the BSEM framework. The aim was to modify the prior specifications for the factor loadings and the covariance matrix of latent factors to explore the diverse effects on multiple Bayesian model fit indices for model fit evaluation and selection.

2 Literature Review

2.1 Bayesian Structural Equation Model

In psychometrics, the CFA model is often utilized as a measurement model within the SEM framework to assess latent constructs' structure (Schreiber et al. 2006; Xiao et al. 2019). The primary focus of this study was to examine the performance of different model fit indices within the BSEM framework. An overview of the CFA model within the BSEM framework was presented. The CFA model with p measurement items and q latent factors is as follows:

$$y_i = v + \Lambda \xi_i + \delta_i, \quad (1)$$

where y_i denotes the data vector ($p \times 1$) for the i -th participant on p measurement items, v is the intercept vector ($p \times 1$), Λ is the factor loading matrix ($p \times q$) representing the relationships between y_i and the latent factor vector ξ_i ($q \times 1$) and δ_i is the measurement error vector ($p \times 1$) for the measurement items. In this case, we assume no correlation exists between the residuals of the measurement items and the latent factors. Furthermore, there is no correlation between the residuals of different measurement items.

Under the frequentist framework, all parameters are considered as constants; only point estimation can be obtained. In BSEM, parameters are treated as random variables, each having its posterior distribution rather than a fixed value. The posterior distribution results from the combination of the prior distribution and data likelihood, written as:

$$P(\theta|Y) = \frac{P(Y|\theta)P(\theta)}{P(Y)}, \quad (2)$$

where $P(\theta|Y)$ is the posterior distribution of parameter θ , $P(Y|\theta)$ is the likelihood function of the data, $P(\theta)$ is the prior distribution, and $P(Y)$ is the marginal distribution of observed data. Muthén and Asparouhov (2012) emphasized that the prior distribution is the crucial element in BSEM, which helps provide a more accurate parameter estimate. Recent studies have demonstrated that BSEM offers greater analysis flexibility than ML estimation methods (Guo et al. 2019). Depaoli (2021) delineates two critical applications of BSEM. Initially, Bayesian methods are exclusively utilized for SEM estimation purposes. Subsequently, Bayesian tools are integrated into the SEM model, resulting in modifications to the model's structure. In such scenarios, applying frequentist methods for model implementation is impractical due to their inherent rigidity, which is crucial for effectively addressing the complexities of such a model.

2.2 Impact of Priors

The prior distribution includes information from previous research and the researchers' beliefs. Based on its information, the prior can be classified into the informative prior, the weak informative prior, and the noninformative prior (or diffuse prior). Researchers determine the hyperparameters for the priors based on the level of certainty associated with the prior information. For example, in the case of a normal distribution prior, researchers can assign a small variance hyperparameter when they have high confidence in the expected parameter values (Muthén and Asparouhov 2012). By combining prior information with observed data, previous research findings can serve as references in theoretical assumptions and directly influence subsequent analysis through prior specifications in the model. Bayesian analysis embodies the accumulation and progression of knowledge, providing a substantial advantage over traditional methods.

While priors provide flexibility and accuracy in parameter estimation, selecting appropriate priors remains one of the major challenges in applying Bayesian methods (Muthén and Asparouhov 2012; van Erp et al. 2018). If priors are incorrectly specified, the resulting posterior distributions will be seriously biased, leading to entirely incorrect conclusions (Baldwin and Fellingham 2013; Depaoli and Clifton 2015; van Erp et al. 2018; Zondervan-Zwijnenburg et al. 2017). Moreover, a review conducted by Smid et al. (2020) indicated that using default noninformative priors in Bayesian estimation for studies with small sample sizes can result in more severe biases than frequentist methods. Biased results can be avoided only by selecting appropriate priors for the parameters. Additionally, the choice of prior distributions for model parameters

substantially impacts parameter estimation. Liu et al. (2016) conducted a study on the influence of inverse Wishart (IW) and separation strategy (SS) priors on parameter estimation in growth curve models. They revealed the distinct roles of these priors in estimating linear and nonlinear growth curve models. The SS priors generally outperformed the IW priors, resulting in smaller biases and higher convergence rates. Similarly, several previous articles have investigated the varying impacts of different priors on parameter estimation (e.g., Depaoli et al. 2021; Pokropek et al. 2020; van Erp et al. 2018).

Currently, research on the impact of prior specification in BSEM mainly focuses on parameter estimation. However, a limited number of studies have investigated the effect of priors on Bayesian model fit indices. Furthermore, there has been insufficient investigation of the newly proposed Bayesian approximation fit indices in Hoofs et al. (2018) and Garnier-Villarreal and Jorgensen (2020).

2.3 Bayesian Model Fit Indices

SEM estimation within the frequentist framework offers a wide range of evaluation indices, including approximate fit indices as well as model selection indices. This provides researchers with valuable information regarding the goodness of fit between the model and the data. However, more research is needed on model fit indices in Bayesian methods. In many application studies, the model fit index in BSEM is limited to posterior predictive p value (PP p value; Winter and Depaoli 2022a, 2022b, 2022c). Moreover, the Bayesian approximate fit indices were introduced relatively recently, and their performance under various model specifications has not been thoroughly investigated (Asparouhov and Muthén 2021). Given the increasing popularity of BSEM, it is crucial to conduct more research on Bayesian model fit indices, which will provide practical recommendations to researchers on the appropriate usage of these indices in various contexts.

2.3.1 Posterior Predictive p value (PP p value)

Posterior Predictive p value (PP p value; Meng 1994) is a model fit index initially introduced in the Bayesian framework. It quantifies the proportion of posterior predictive discrepancy statistics exceeding the discrepancy statistics of the observed data (Cain and Zhang 2019). Typically, a PP p value close to 0.5 indicates a good model fit. In contrast, a PP p value close to 0 indicates poor model fit. Previous studies have shown that the PP p value can be influenced by varying prior settings (e.g., Cain and Zhang 2019; Liang 2020; Winter and Depaoli 2022b). In particular, when prior settings deviate from the true parameter values, the PP p value of the model can be substantially reduced. In small samples, the PP p value may reject the correctly specified model (Winter and Depaoli 2022b).

2.3.2 Deviance Information Criterion (DIC)

The Bayesian information criterion (BIC; Schwarz 1978) and the deviance information criterion (DIC; Spiegelhalter et al. 2002) are the most widely used model selection indices in Bayesian models. Asparouhov et al. (2015) suggested that using DIC as the evaluative index for model selection performs better in BSEM compared to BIC. Previous research has also indicated that DIC was the most sensitive among other Bayesian model selection indices (e.g., Liang and Luo 2020; Pokropek 2020). Therefore, DIC was selected as the particular Bayesian model selection index to investigate.

Previous studies have explored the performance of DIC under different priors and have shown that different prior settings exert an influence on DIC (e.g., Cain and Zhang 2019; Liu et al. 2022; Pokropek et al. 2020; Winter and Depaoli 2022b).

2.3.3 Bayesian Approximate Fit Indices

Hoofs et al. (2018) first proposed a Bayesian variant of the root mean square error of approximation (RMSEA), known as BRMSEA. Garnier-Villarre and Jorgensen (2020) subsequently adapted fit indices from the frequentist framework to the Bayesian framework. They developed additional Bayesian approximation fit indices, such as the Bayesian variant of comparative fit index (BCFI) and Tucker Lewis index (BTLI). Bayesian approximate fit indices, including BRMSEA, BCFI, and BTLI, have been implemented in the *Mplus* software (Asparouhov and Muthén 2021). The details for calculating BRMSEA, BCFI, and BTLI are in the Appendix.

There is a lack of comprehensive research on the impact of priors on Bayesian approximate fit indices. Winter and Depaoli (2022b) showed that when deviating from the true parameter values, the prior exerted the most substantial impact on BRMSEA in latent growth models, followed by *PPp* value and the least notable impact on BCFI and BTLI. However, regarding model selection, the adverse effects of incorrect priors had little impact on selecting the correct model based on Bayesian approximate fit indices. Liang's (2020) study focused on normal priors with small variance hyperparameters for cross-loadings and found that BCFI, BTLI, and BRMSEA had low sensitivity to prior information. Edwards and Konold (2022) focused on priors of factor loadings and how they affected the Bayesian approximate fit indices and mentioned that the influence of priors on the latent factor covariance matrix should also be investigated. The sensitivity of Bayesian approximate fit indices to priors has not been extensively investigated, and the results of existing studies have yielded conflicting conclusions.

2.4 The Present Study

Previous research primarily focuses on the effects of priors on parameter estimation, with some studies also investigating the impact of priors on model evaluation indices. However, there is a need for a more comprehensive exploration of the

Bayesian approximate fit indices, and inconsistent conclusions have been reported. To enhance our understanding of the performance of Bayesian approximate fit indices under different priors, two simulation studies and one empirical study were conducted to evaluate the impact of different priors in BSEM on DIC, PPp values, and Bayesian approximate fit indices. It included assessing the performance of model fit and the ability to select the appropriate model.

3 Simulation Study 1

3.1 Design

3.1.1 Data Generation

Based on the study conducted by Garnier-Villarreal and Jorgensen (2020), a three-factor CFA model was employed as the population model, as shown in Fig. 1. Each factor consisted of five items with no cross-loadings. Factor 1 encompassed items 1 to 5, while factor 2 comprised items 6 to 10, and factor 3 consisted of items 11 to 15. The variance for each factor was set to 1.0, and the correlation between factors was set to 0.3. To ensure model identification, the first factor loading in each factor was fixed at 1.0, while the rest factor loadings were set to 0.8. All residual variances were set to 0.36.

According to Muthén and Asparouhov (2021), the approximate fit indices are suggested to be used with large sample sizes, specifically when the sample size exceeds 100 or even 200. Kyriazos (2018) summarized previous studies by finding that a sample size of 500 was sufficient to provide unbiased estimates for most applications. Hence, to examine the impact of sample size, we utilized five different sample size levels: 100, 200, 300, 400, and 500. *Mplus* version 8.6 was used to generate data, and for each sample size, 500 replications were generated.

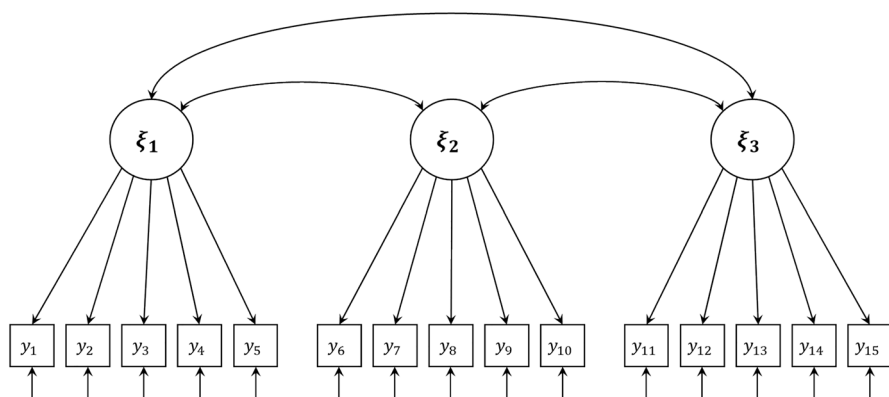


Fig. 1 Data generation and fitting model in Simulation Study 1 (M1)

Table 1 Prior settings for different parameters

Abbreviation	Factor loadings	Variance–covariance matrix of latent factors	
		The entire matrix	Separation strategy (diagonal / off-diagonal)
	$\lambda_y \sim N\left[\mu_{\lambda_y}, \sigma_{\lambda_y}^2\right]$	$\Phi_{\xi} \sim IW[\Psi, \nu]$	$\sigma_{\xi}^2 \sim IG\left[a_{\sigma_{\xi}^2}, b_{\sigma_{\xi}^2}\right] / \rho_{\xi} \sim U[l, u]$
N1	$N(0, \infty)$		
N2	$N(0, 10)$		
N3	$N(0, 0.01)$		
IW1		$IW(0, -4)$	
IW2		$IW(I_{3 \times 3}, 3)$	
IW3		$IW(I_{3 \times 3}, 9)$	
IW4		$IW(I_{3 \times 3}, 25)$	
SS1			$IG(-1, 0) / U[-1, 1]$
SS2			$IG(0.001, 0.001) / U[-1, 1]$
SS3			$IG(0.01, 0.01) / U[-1, 1]$
SS4			$IG(0.1, 0.1) / U[-1, 1]$

The priors with underscore represent the default noninformative priors in *Mplus*

3.1.2 Prior Settings

The model estimated in this study was consistent with the data generation model, as depicted in Fig. 1. In Simulation Study 1, we used 3 (factor loadings) $\times$ 8 (latent factor matrix) = 24 different prior settings (see Table 1).

3.1.2.1 Factor Loadings For primary factor loadings (without cross-loadings in this study), several normal distribution priors with mean 0 and different variances were assigned. The $N(0, \infty)$ is the default noninformative normal prior for factor loadings in *Mplus* (Muthén and Asparouhov 2012). For the normal prior hyperparameters, a smaller variance implies greater certainty for the parameter. And a larger variance indicates less available information and greater uncertainty in the parameter value. Priors were selected as $N(0, 0.01)$, $N(0, 10)$, and $N(0, \infty)$ for factor loadings with varying levels of information.

3.1.2.2 Variance–Covariance Matrix of Latent Factors Two methods were used to specify priors for the variance–covariance matrix of latent factors: setting multivariate priors for multiple matrix elements or setting univariate priors for individual components separately.

When specifying multivariate priors for the entire variance–covariance matrix, we utilized the default noninformative prior by *Mplus* and different informative IW priors. The default noninformative prior in *Mplus* is $IW(0, -q - 1)$, with q representing the number of latent factors in the model (Muthén and Asparouhov 2012). For the IW prior hyperparameters, larger degrees of freedom (df) imply smaller variances, indicating a higher degree of informativeness for the prior. $IW(0, -q - 1)$,

$IW(I, q)$, $IW(I, q + 6)$, and $IW(I, q + 22)$ were selected as multivariate priors. The identity matrix, denoted as I , was employed as the scale matrix for the IW distribution, with diagonal elements set to 1 and off-diagonal elements to 0. Since the model used in this study includes three latent factors, the selected priors can be represented as $IW(0, -4)$, $IW(I, 3)$, $IW(I, 9)$, and $IW(I, 25)$.

Different informative SS priors were employed when specifying univariate priors for the diagonal and off-diagonal elements of the matrix individually. The inverse Gamma (IG) priors were specified for the diagonal elements. The $IG(-1, 0)$ serves as the default noninformative prior for estimating variance parameters in *Mplus* (Asparouhov and Muthén 2010). However, there is a lack of clear explanation regarding the specification of the shape and scale hyperparameters for the IG priors, and few studies have examined the impact of these priors on fit indices in BSEM (van Erp et al. 2018). According to van Erp et al. (2018), we selected $IG(-1, 0)$, $IG(0.001, 0.001)$, $IG(0.01, 0.01)$, and $IG(0.1, 0.1)$ as priors for the diagonal elements of the variance–covariance matrix of the latent factors. The uniform distribution prior $U[-1, 1]$ was specified as the prior for the off-diagonal elements of the variance–covariance matrix.

3.1.2.3 Residual Variances Under all conditions, the residual variances were assigned with the default noninformative prior in *Mplus*, which is the $IG(-1, 0)$.

3.1.3 Evaluation Criteria

We used the *MplusAutomation* package (Hallquist and Wiley 2018) in R version 4.0.4 to call *Mplus* version 8.6 to conduct Bayesian estimation. A total of 5 (sample size levels) $\times 24$ (prior conditions) = 120 conditions were examined. Two MCMC chains were generated for analysis, with a minimum of 10,000 and a maximum of 50,000 iterations.

3.1.3.1 Model Convergence Rate We used potential scale reduction (PSR) to evaluate the convergence of the model. We utilized the default PSR value of 1.05 in *Mplus* version 8.6 to determine whether the model had achieved convergence. In particular, a model is considered to have converged when the PSR is less than 1.05.

3.1.3.2 Model Fit We used the cutoff values for the Bayesian model fit indices to assess the model's fit. Although in Bayesian estimation, it is possible to obtain 90% credible intervals for the fit indices, previous studies have indicated that various factors can influence the results of credible intervals and may lack practical significance (Winter and Depaoli 2022a 2022b). Therefore, we focused on the cutoff values for Bayesian model fit indices in this study.

A PPp value of approximately 0.5 suggests a good fit of the model (Asparouhov and Muthén 2012). However, Asparouhov and Muthén (2010) recommended using a cutoff value of $PPp = 0.05$ to determine an acceptable model fit, as this cutoff value balanced the sensitivity of misspecification and the Type I error rate. Therefore,

a cutoff value of $PPp=0.05$ was utilized. If the PPp value is larger than 0.05, the model is considered acceptable.

We employed the cutoff values for the approximative fit indices under the frequentist framework as the criteria for evaluating the fit of the model within the Bayesian framework, as the Bayesian approximative fit indices demonstrate comparable performance to those approximative fit indices (Asparouhov and Muthén 2021). Based on the cutoff values proposed by Hu and Bentler (1999), we used $BRMSEA=0.06$, $BTLI=0.95$, and $BCFI=0.95$ as the cutoff values. The model is considered a good fit if $BCFI$ and $BTLI$ are larger than 0.95. If $BRMSEA$ is less than 0.06, the model is also considered a good fit.

3.2 Results

Convergence was achieved for all models under all conditions. Due to the similarity between the $BCFI$ and BTL results, the interpretation focused primarily on the results obtained from $BCFI$. Box plots (Figs. 2, 3 and 4) were used to present the results of $BCFI$, PPp values, and $BRMSEA$. The data regarding specific model fit indices and related images for all conditions in Simulation Study 1 have been uploaded to the OSF of this study. They can be accessed through the following link (https://osf.io/fvkwh/?view_only=9592379e0f5b442e9260b61c3b6a5ce).

3.2.1 Impacts of Priors on $BCFI$

The $BCFI$ values obtained under different conditions are presented in Fig. 2. In all previous conditions, when the sample size was increased, there was an increase in the value of $BCFI$, indicating an increase in the likelihood of obtaining a good model fit. For sample sizes of 400 and 500, despite variations in $BCFI$ among different priors, the $BCFI$ indicated a good model fit under all prior conditions.

Different parameters and prior settings have varying effects on the $BCFI$ for prior conditions. For factor loadings, when specifying the normal prior $N(0, 0.01)$, there was a substantial decrease in $BCFI$ with a sample size of 100, below the cutoff value (0.95), indicating poor model fit. Despite increasing the sample size to $N=200$ and $N=300$, cases still existed where $BCFI$ was lower than the predetermined cutoff value. Only when the sample exceeded 400 did all $BCFI$ values become larger than the predetermined cutoff value. The $BCFI$ performed similarly when priors were set to $N(0, 10)$ or $N(0, \infty)$: in sample sizes larger than 200, weakly informative and noninformative priors had little impact on $BCFI$; with an increasing sample size, the influence of priors with the highest information level on $BCFI$ gradually diminishes.

We discussed the results of different priors under two strategies for the prior setting of the latent factor variance–covariance matrix. When setting the IW priors (Fig. 2A), various levels of information in the prior affected the $BCFI$. For a sample size of 100, and the priors for factor loadings setting as $N(0, 10)$ and $N(0, \infty)$, the $BCFI$ values under the $IW(I, 25)$ condition were markedly lower than under other IW priors with half of the $BCFI$ indicating poor model fit. When setting the priors for factor loadings as $N(0, 0.01)$, all $BCFI$ values under different IW priors showed

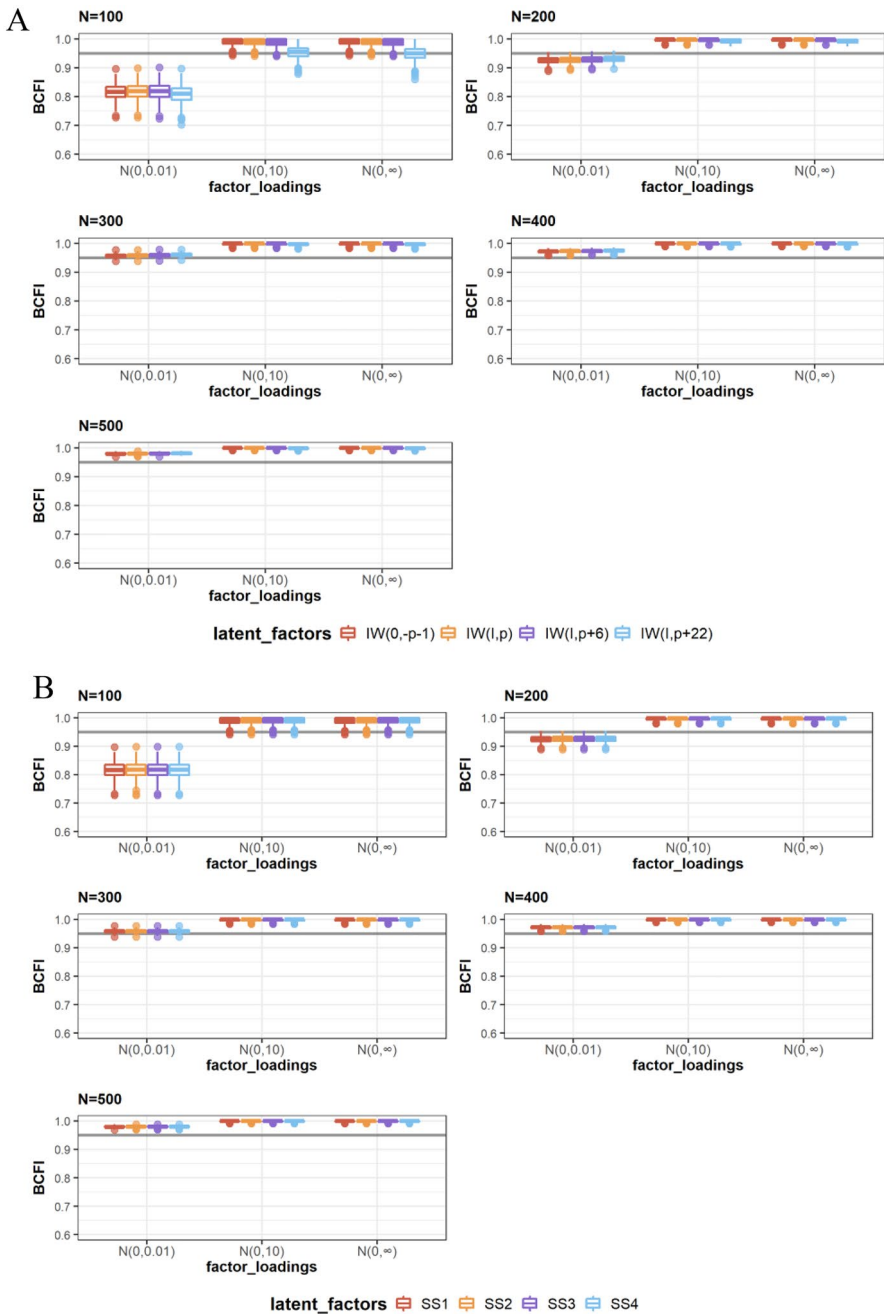
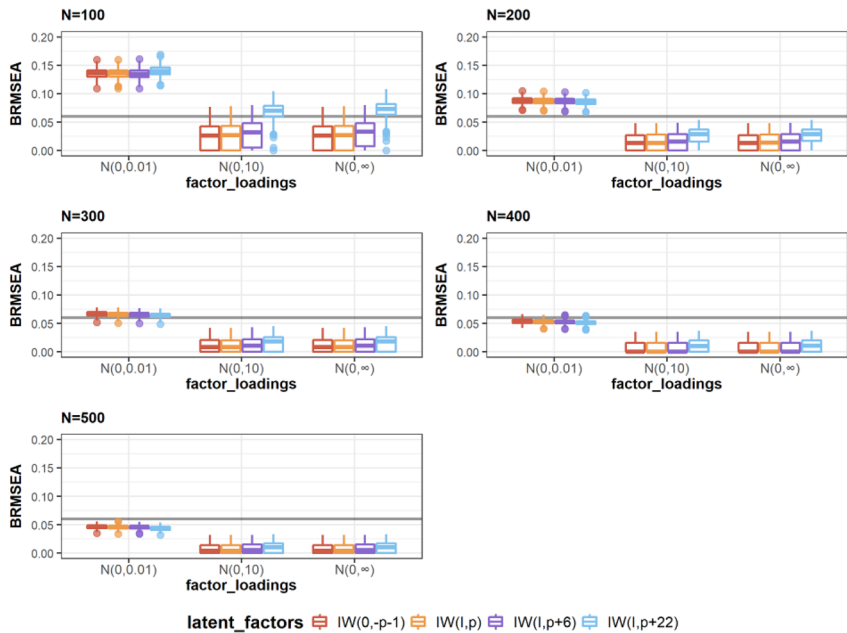


Fig. 2 Results of BCFI in Simulation Study 1. *Note.* The solid gray line represents the cutoff value BCFI=0.95. Panel A: The inverse Wishart (IW) priors were specified for variance–covariance matrix of latent factors. Panel B: The separation strategy (SS) priors were specified for variance–covariance matrix of latent factors

A



B

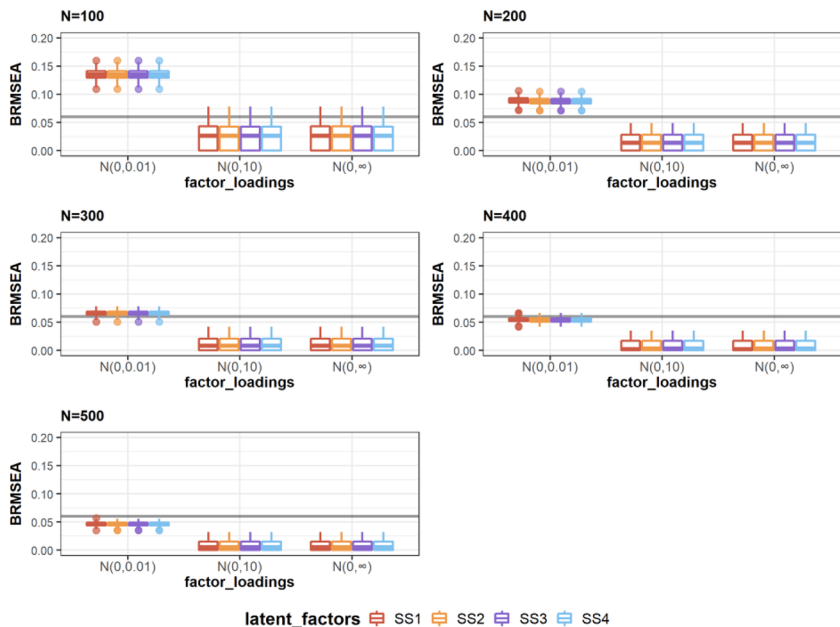


Fig. 3 Results of BRMSEA in Simulation Study 1. *Note.* The solid gray line represents the cutoff value BRMSEA=0.06. Panel A: The inverse Wishart (IW) priors were specified for variance–covariance matrix of latent factors. Panel B: The separation strategy (SS) priors were specified for variance–covariance matrix of latent factors

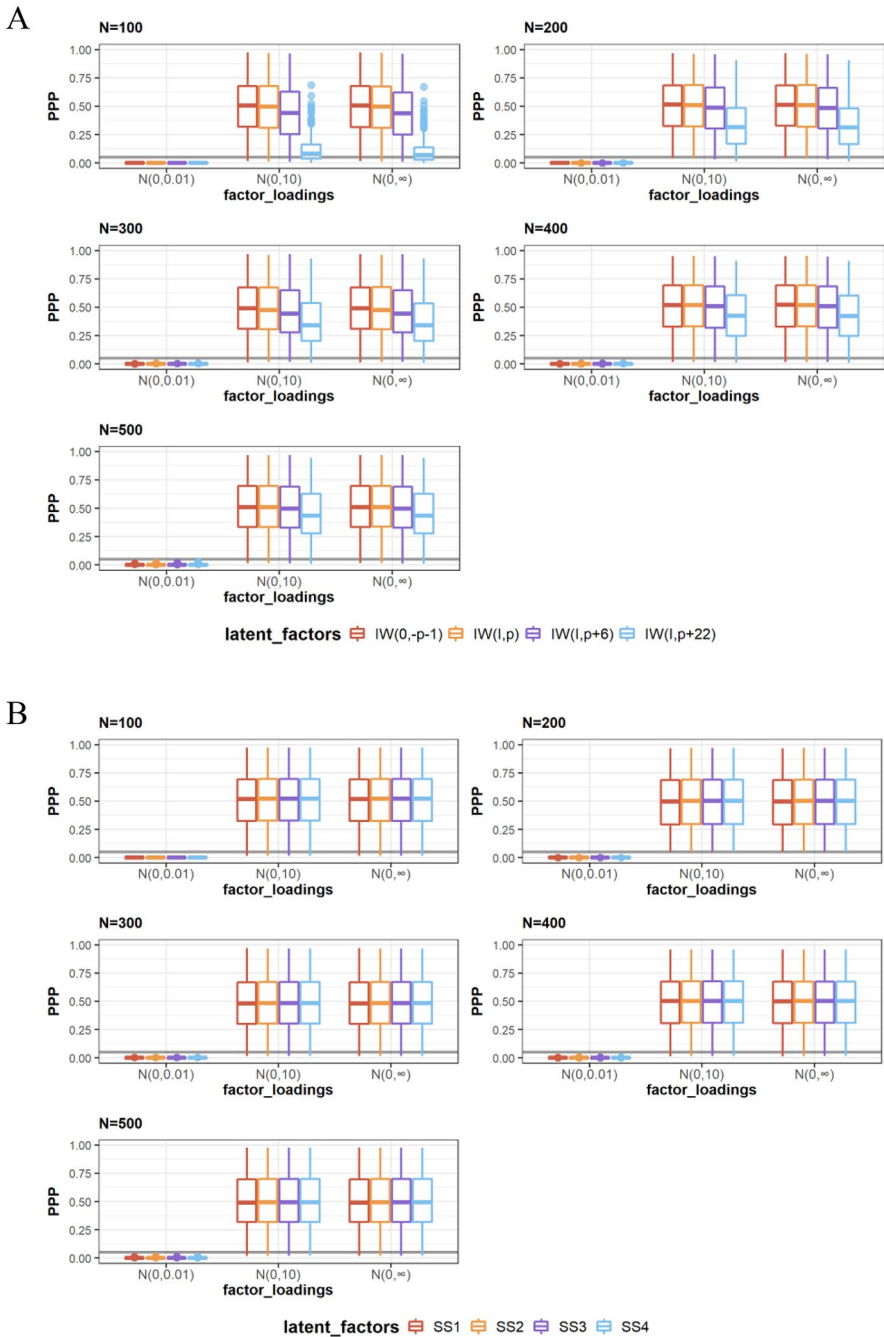


Fig. 4 Results of PPP value in Simulation Study 1. *Note.* The solid gray line represents the cutoff value $PPP=0.05$. Panel A: The inverse Wishart (IW) priors were specified for variance–covariance matrix of latent factors. Panel B: The separation strategy (SS) priors were specified for variance–covariance matrix of latent factors

similar performance, indicating poor model fit. With a sample size of 200 or more, there was almost no difference in BCFI under different IW priors, and all BCFI values substantially exceeded the cutoff value. Therefore, BCFI showed high sensitivity to IW priors with the highest information level ($df=25$). However, this sensitivity was less pronounced compared to the sensitivity to the factor loading priors. When univariate priors were specified separately (Fig. 2B), different shape and scale hyperparameters specified in the SS prior did not substantially affect BCFI.

3.2.2 Impacts of Priors on BRMSEA

The BRMSEA values obtained under different conditions are displayed in Fig. 3. Compared to BCFI, the impact of sample size on BRMSEA was less pronounced. A gradual decrease in BRMSEA was observed as the sample size increased, indicating improved model fit. However, only when the sample size reached 500 did BRMSEA become smaller than the cutoff value under all prior conditions.

The impact of the prior setting of factor loadings on BRMSEA was more notable than that of BCFI. For sample sizes less than or equal to 300, the model with prior of $N(0, 0.01)$ for factor loadings exhibited a poor model fit, as evidenced by BRMSEA. For a sample size of 400, BRMSEA fluctuated around the cutoff value, with some values exceeding the cutoff value. This finding demonstrated that BRMSEA was highly sensitive to the small variance in the normal priors, even with a large sample size. An erroneous normal prior with a small variance can still lead to BRMSEA indicating a poor model fit.

When setting the IW priors for the entire latent factor variance–covariance matrix (Fig. 3A), with a sample size of 100 and factor loading priors of $N(0, 10)$ and $N(0, \infty)$, it was found that under the $IW(I, 25)$ condition more than half of the models exhibited poor fit based on BRMSEA values. The median of BRMSEA values under the $IW(I, 9)$ condition was slightly higher than those of the other two IW priors but did not surpass the cutoff value, suggesting that most models exhibited good model fits. It can be seen that BRMSEA also showed some sensitivity to $IW(I, 9)$. The results for the SS prior setting were consistent with BCFI results (Fig. 3B), and the impact of different scale and shape values of the SS prior on BRMSEA remained consistent.

3.2.3 Impacts of Priors on PPp value

Figure 4 displays the PPp-values for different conditions. The effect of varying sample sizes on PPp-values to priors was smaller than those on BCFI and BRMSEA. Except for the case where the priors for factor loadings were $N(0, 10)$ or $N(0, \infty)$, and the priors for the latent factor matrix were $IW(I, 25)$, the influence of the sample size on PPp-values was negligible.

The PPp-values were most affected by the priors of factor loadings among all fit indices considered. Regardless of sample size, setting $N(0, 0.01)$ for factor loadings resulted in the PPp-values approaching 0. Although increasing the sample size caused fluctuations in the PPp-values, none exceeded the cutoff value 0.05.

When setting the IW prior for the entire latent factor matrix (Fig. 4A), there was still little difference in the impact between setting $IW(0, -4)$ and $IW(I, 3)$. When the prior was set to $IW(I, 9)$, there was a slight decrease in the PPp -values, but they remained above the cutoff value. However, when the prior was set to $IW(I, 25)$ with the factor loading priors of $N(0, 10)$ or $N(0, \infty)$, there was a decrease in the PPp -values. When the sample size was 100, nearly half of the models had PPp -values below 0.05, suggesting inadequate model fit. When a sample size exceeded 200, the PPp -values, while lower than other IW priors, exceeded the cutoff value, indicating a well-fitting model. When setting the SS prior (Fig. 4B), there remained no substantial difference in the impact of different shape and scale hyperparameters on the PPp -values.

4 Summary

Both sample size and prior settings had different effects on assessing model fit using Bayesian model fit indices. Models with a sample size below 200 tended to exhibit poorer model fit, as the BCFI, BTLI, BRMSEA, and PPp values indicated, mainly when informative prior settings were applied. However, it can be seen that among all Bayesian model fit indices, the PPp -values are most affected by the factor loading priors. When the sample size was larger than 200, BCFI, BTLI, and BRMSEA showed reduced sensitivity to different priors, thereby minimizing the risk of incorrectly rejecting the correct model. As a result, Bayesian approximate fit indices may be more suitable for model evaluation compared to PPp -values.

Simulation Study 1 focused solely on investigating how sample size and priors influenced the performance of Bayesian model fit indices in assessing model fit. Nonetheless, there is still a need for further investigation into how these conditions affect the ability of the Bayesian model fit index to select the correct model from a range of different models. Therefore, Simulation Study 2 was conducted to investigate the model selection capabilities of BCFI, BTLI, BRMSEA, and DIC by new data generation and fitting models. It aimed to examine their performance under different sample sizes and priors.

5 Simulation Study 2

5.1 Design

5.1.1 Data Generation

We utilized the data generation model employed in Simulation Study 1, which did not include residual correlations and cross-loadings, with population parameters consistent with Simulation Study 1 (Fig. 1), labeled M1. Based on M1, we introduced two additional data generation models, M2 and M3, which included cross-loadings or residual correlations, respectively (see Fig. 5). In M2, two cross-loadings were added on the base of M1, with one from factor 2 to item 4 and another

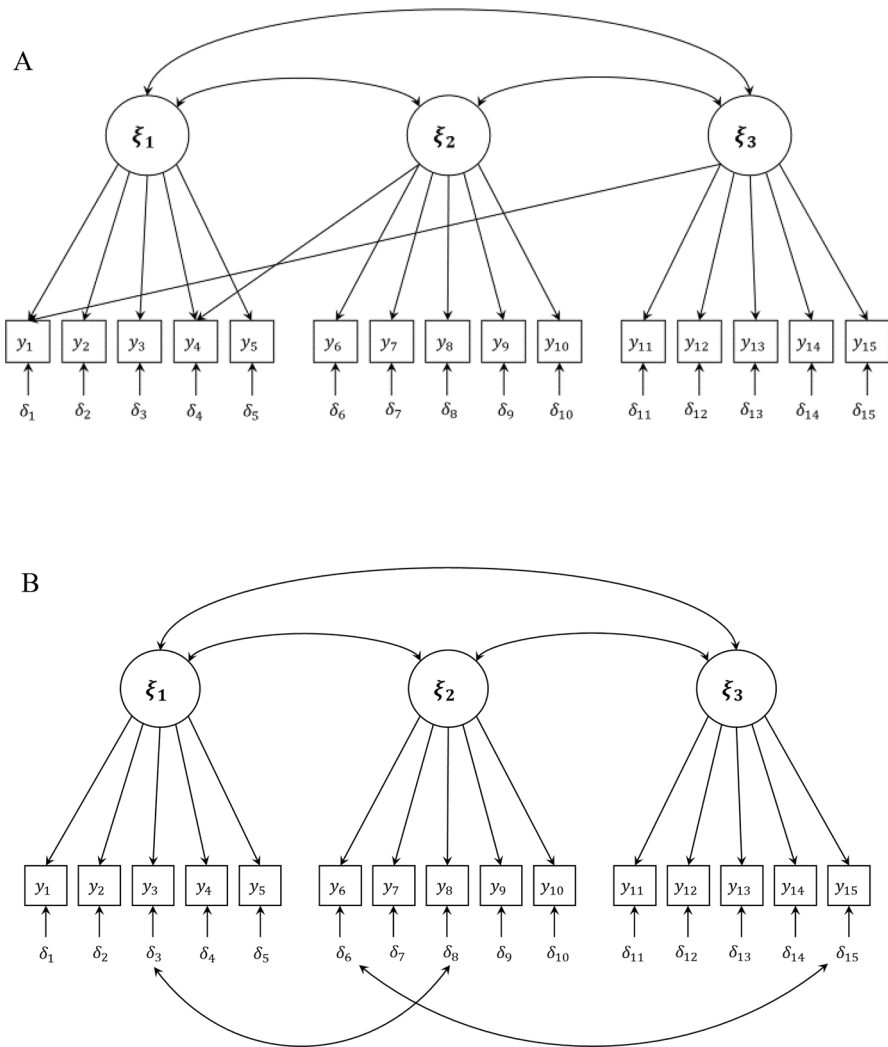


Fig. 5 Newly added data generation and fitting models in Simulation Study 2 (M2 and M3)

from factor 3 to item 1. The cross-loading in M2 was set to 0.2. Furthermore, M3 was derived from M1, including residual correlations of 0.3 between item 3 and item 8 and item 6 and item 15. The population values of other parameters of M2 and M3 were consistent with those of M1.

The sample size levels were the same as those used in Simulation Study 1, which are 100, 200, 300, 400, and 500, respectively. *Mplus* version 8.6 was used to generate the data. The data generation conditions included three different population models and five sample size levels, resulting in 15 (3×5) conditions. For each condition, 500 replications were generated.

5.1.2 Fitting Models

The data generated by three different data generation models were analyzed using both correct and incorrect models. For the data generated by M1, the proper model was M1, whereas the incorrect models were M2 and M3, which included cross-loadings or residual correlations absent in the population model. In this case, M2 and M3 were more complex than the population model, also known as the overfitted model.

For the data generated by M2, the correct model was M2, while the incorrect model was M1, which ignored the cross-loadings. Similarly, when analyzing the data generated by M3, the correct model was M3, whereas the incorrect model was M1, which failed to consider the residual correlations. In both cases, the incorrect models exhibited a simpler structure than the population model, also called the misspecified model.

In Simulation Study 2, seven combinations existed between the data generation and fitting models.

5.1.3 Prior Settings

Similar to Simulation Study 1, we employed $3 \text{ (factor loadings)} \times 8 \text{ (latent factor matrix)} = 24$ different prior settings in Simulation Study 2, as shown in Table 1. The same priors were specified for both primary factor loadings and cross-loadings (from factor 2 to item 4 and factor 3 to item 1 in M2). Under all conditions, the residual variances were assigned with the default noninformative prior in *Mplus*, the $IG(-1, 0)$.

5.1.4 Evaluation Criteria

Bayesian estimation was conducted using the *MplusAutomation* package (Hallquist and Wiley 2018) in R version 4.0.4 to call *Mplus* version 8.6. A total of $5 \text{ (sample size levels)} \times 24 \text{ (prior conditions)} \times 7 \text{ (model combinations)} = 840$ conditions were examined. Two MCMC chains were generated, with a minimum of 10,000 and a maximum of 50,000 iterations.

5.1.4.1 Model Convergence Rate We utilized the default PSR value of 1.05 of PSR in *Mplus* version 8.6 to determine whether the model converged. In particular, a model is considered to have converged when the PSR is less than 1.05.

5.1.4.2 Model Selection The model with a smaller DIC better fits the data in comparison. Generally, the DIC (ΔDIC) difference between the two models is computed. Choosing $\Delta DIC = 7$ as the cutoff value reduces the likelihood of selecting the incorrect model (Cain and Zhang 2019). Consequently, we used $\Delta DIC = 7$ as the cutoff value for model selection. If ΔDIC is greater than 7 or less than -7 , the

model with the smaller DIC is selected. If ΔDIC is between -7 and 7 , DIC does not provide a definitive preference between the two models.

For the Bayesian approximate fit indices, we also used the cutoff values of approximate fit indices in the frequentist framework as the criteria for model selection (Asparouhov and Muthén 2021). We use $\Delta\text{BCFI}=0.005$, $\Delta\text{BTLI}=0.005$, and $\Delta\text{BRMSEA}=0.010$ as the cutoff values to indicate significant model differences between models (Chen 2007; Sokolov 2019). If ΔBCFI and ΔBTLI exceed 0.005 or fall below -0.005 , the model with higher values of BCFI and BTLI will be selected. When ΔBCFI and ΔBTLI are within the range of -0.005 and 0.005 , they do not make a definitive preference for either of the two models. When ΔBRMSEA is greater than 0.010 or less than -0.010 , the model with the lower BRMSEA will be selected. If ΔBRMSEA lies within the range of -0.010 and 0.010 , BRMSEA does not make a clear choice between the two models.

5.2 Results

Across 840 conditions, the convergence rates exceeded 99.5%. Only the results obtained from the converged models will be discussed below.

Two scenarios emerged from the model selection process: (1) The data generation model (population model) was simple, while the incorrect data fitting model was complex. (2) Conversely, the data generation model was complex, while the incorrect data fitting model was simple. The performance of Bayesian approximate fit indices and DIC in model selection was separately discussed for each scenario.

The values of the model fit indices for each model, the differences in the fit index, the model selection rates in specific conditions, and graphs have been uploaded to the OSF. They can be accessed through the following link: (<https://osf.io/fvkwk/?view=9592379e0f5b442e926f0b61c3b6a5ce>).

5.2.1 Complex Incorrect (overfitted) Model

5.2.1.1 The Correct Model M1 with the Incorrect Model M2 In analyzing the data generated from M1 using M2, the primary focus of our study was to determine if Bayesian model fit indices can accurately select the correct (M1) rather than the incorrect (M2). We also compared the differences in model selection across different fit indexes (see Fig. 6).

In most prior conditions and when the sample size was large ($n \geq 200$), almost all Bayesian model fit indices failed to distinguish between the correct and the incorrect models. However, model selection was influenced by different prior settings for sample sizes of $n = 100, 400$, and 500 .

When the IW prior was assigned to the variance–covariance matrix of the latent factor (Fig. 6A), its effect on model selection, as indicated by Bayesian fit indices, was most pronounced for a sample size of 100 . Among the various prior settings for factor loadings, it was found that only $N(0,0.01)$ (N3) had a distinguishable impact on the Bayesian fit indices. Regarding BRMSEA, the $N(0,$

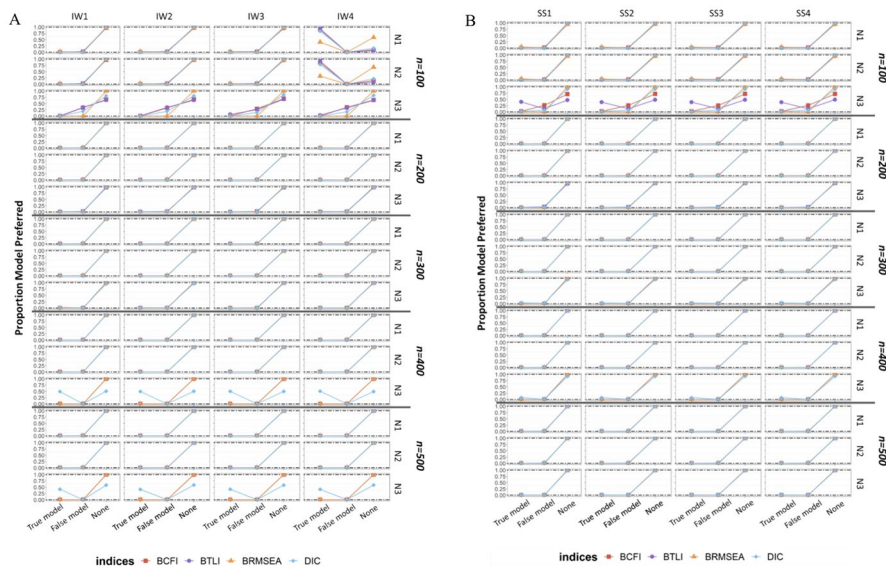


Fig. 6 Model selection rates between the correct model M1 and the incorrect model M2. *Note.* In the horizontal axis, "True model" represents selecting the correct model (M1); "False model" represents selecting the incorrect model (M2); "None" indicates that the fit indices did not choose between the correct and incorrect models. The abbreviations N1–N3 represented $N(0, \infty)$, $N(0, 10)$, and $N(0, 0.01)$; IW1–IW4 represented IW(0, -4), IW(I , 3), IW(I , 9), and IW(I , 25); and SS1–SS4 represented IG(-1, 0), IG(0.001, 0.001), IG(0.01, 0.01), and IG(0.1, 0.1). Bold black lines separated different sample sizes represented by " n "

0.01) distribution demonstrated a similar effect to other normal priors, showing an inability to differentiate between the correct and incorrect models across all sample sizes. However, when using the $N(0, 0.01)$ prior and under the sample size of 100, BCFI, BTLI, and DIC exhibited error model selection rates of approximately 25%, with BCFI and BTLI showing a higher tendency to select the incorrect model compared to DIC.

For the factor variance–covariance matrix, only IW(I , 25) (IW4) had a substantial but relatively minor impact compared to the effects of factor loading priors. Therefore, IW(I , 25) demonstrated a distinct effect compared to other IW priors only when the priors for the factor loadings were specified as $N(0, 10)$ or $N(0, \infty)$. Under the situations, BCFI, BTLI, and DIC demonstrated the capability to accurately identify the correct model, with the proportion of selecting the incorrect model or failing to make a selection close to 0. However, BRMSEA exhibited relatively poor performance. Although it could not select the incorrect model, it only chose the correct one in less than 50% of the cases and often failed to choose between the two models.

Additionally, when the sample size was 400 or greater, DIC accurately identified the correct model in approximately 50% of the cases when factor loading prior was specified as $N(0, 0.01)$. However, this result was not observed when the SS prior was specified for the factor variance–covariance matrix. The results suggested that, in the case of a large sample, DIC was influenced by both the normal

prior for the factor loadings and the IW prior for the latent factor matrix, showing a tendency for selecting a more parsimonious model.

When specifying the SS prior for the variance–covariance matrix of latent factor (Fig. 6B), it was observed that different SS priors had almost the same effects on the Bayesian model fit indices. The prior distributions of the factor loadings also had the most substantial impact on model selection when $n=100$. In the cases of the $N(0, 10)$ and $N(0, \infty)$ (referred to as N1 and N2, respectively), the Bayesian fit indices failed to differentiate between the two models. Under the $N(0, 0.01)$ (N3), different fit indices showed different preferences for model selection, which contrasted with the case of the IW prior. BTLI correctly selected the model in approximately 50% of cases. BCFI exhibited a similar pattern to that shown in Fig. 6A, with approximately 25% of cases selecting the incorrect model. BRMSEA also exhibited consistency with Fig. 6A but could not differentiate between the two models. DIC changed, with a reduced tendency to choose the incorrect model compared to Fig. 6A. It showed a similar pattern to BRMSEA but struggled to differentiate between the two models.

5.2.1.2 The Correct Model M1 with the Incorrect Model M3 While analyzing the data generated from M1 using M3, we examined whether Bayesian model fit indices can

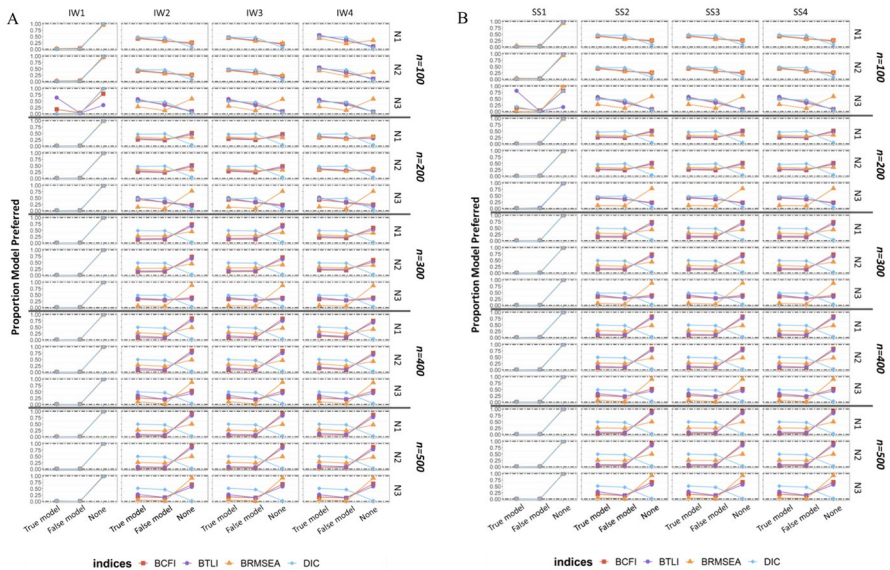


Fig. 7 Model selection rates between the correct model M1 and the incorrect model M3. *Note.* In the horizontal axis, "True model" represents selecting the correct model (M1); "False model" represents selecting the incorrect model (M3); "None" indicates that the fit indices did not choose between the correct and incorrect models. The abbreviations N1–N3 represented $N(0, \infty)$, $N(0, 10)$, and $N(0, 0.01)$; IW1–IW4 represented $IW(0, -4)$, $IW(I, 3)$, $IW(I, 9)$, and $IW(I, 25)$; and SS1–SS4 represented $IG(-1, 0)$, $IG(0.001, 0.001)$, $IG(0.01, 0.01)$, and $IG(0.1, 0.1)$. Bold black lines separated different sample sizes represented by " n ="

correctly select the model M1 instead of the incorrect model M3. We also compared the differences in model selection among different fit indices (see Fig. 7).

Overall, the impact of different priors on the selection between M1 and M3 was greater than the impact on that between M1 and M2. As the sample size increased, the effect of different priors gradually decreased. The sample size exerted the most substantial influence on BCFI and BTLI, with BRMSEA following behind, while the impact on DIC was the smallest. In the case of the smallest sample size and informative priors assigned for the latent factor matrix, BCFI, BTLI, and BRMSEA exhibited similar performance to DIC, with a selection rate of approximately 50% for the correct model but a slightly lower rate for the incorrect model when compared to DIC. As the sample size increased, the likelihood that BCFI, BTLI, and BRMSEA could not distinguish between the two models also increased. In contrast, DIC maintained a selection rate of approximately 50% for both the correct and incorrect models.

The results of setting the IW and SS priors for the latent factor matrix were highly comparable. All informative priors yielded distinct effects compared to the noninformative priors. However, the differences between different levels of informative priors were not substantial, except for the case when the sample size was 100. In this instance, IW(I , 25) (IW4) showed subtle differences in the conditions with lower degrees of freedom (IW2 and IW3). For other conditions, the impact of the SS priors closely mirrored that of the IW priors; therefore, we discussed the results obtained from setting the IW prior as representative for further analysis (Fig. 7A).

The impact of $N(0, 0.01)$ (N3) on the model fit indices was still most substantial for the priors for the factor loading. However, the influence of the latent factor matrix priors also played a role in shaping this impact. When the prior was set to $N(0, 0.01)$ and a noninformative IW prior setting, BCFI and BTLI were affected at $n=100$. Specifically, BTLI displayed an increased correct model selection rate of about 60%, while BCFI exhibited a correct model selection rate of around 20%. This resulted in a reduction in the proportion of cases where no choice was made. Under the $N(0, 0.01)$ and informative IW priors setting, BRMSEA was the most affected, resulting in a substantial increase in the proportion of cases where no model selection was made across all sample sizes. BCFI and BTLI exhibited an increase in the selection rate for correct and incorrect models, although this effect decreased with increase in sample size. DIC, on the other hand, was hardly affected.

Compared to Fig. 6A, the influence of the latent factor matrix priors was more pronounced when M1 was the correct model and M3 was incorrect. There were noticeable differences between the noninformative IW prior and the informative IW priors. Under the noninformative IW prior, none of the fit indices select the incorrect model. For sample sizes of 200 or larger, none of the indices could distinguish between the two models. When the informative IW priors were used, DIC was the most affected, with the incorrect selection rate and the correct selection rate being around 50% across all sample sizes and factor loading prior conditions. BCFI, BTLI, and BRMSEA were most affected by the sample size of 100, with the correct model being selected at about 50% and the incorrect model being selected at a rate of about 30%. As the sample size increased, the impact of the informative IW prior to BRMSEA was greater than that of BCFI and BTLI, but all the impacts gradually

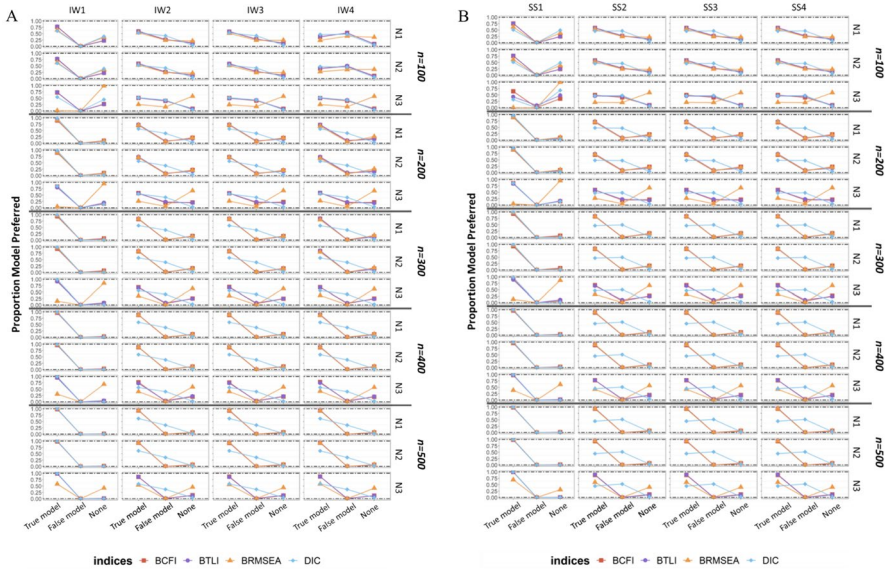


Fig. 8 Model selection rates between the correct model M2 and the incorrect model M1. *Note.* In the horizontal axis, "True model" represents selecting the correct model (M2); "False model" represents selecting the incorrect model (M1); "None" indicates that the fit indices did not choose between the correct and incorrect models. The abbreviations N1–N3 represented $N(0, \infty)$, $N(0, 10)$, and $N(0, 0.01)$; IW1–IW4 represented $IW(0, -4)$, $IW(I, 3)$, $IW(I, 9)$, and $IW(I, 25)$; and SS1–SS4 represented $IG(-1, 0)$, $IG(0.001, 0.001)$, $IG(0.01, 0.01)$, and $IG(0.1, 0.1)$. Bold black lines separated different sample sizes represented by " n "

decreased. The Bayesian approximate fit indices could not distinguish between the two models in large samples.

5.2.2 Parsimonious Incorrect (Misspecified) Model

5.2.2.1 The Correct Model M2 with the Incorrect Model M1 When analyzing the data generated by M2 using M1, which ignored two cross-loadings, we aimed to investigate whether the Bayesian model fit indices effectively identify the correct model (M2) instead of erroneously selecting the model (M1). Additionally, we compared the differences in model selection across different fit indices (see Fig. 8).

The selection rate for the correct model, measured by BCFI, BTLI, and BRMSEA, increased as the sample size increased. Concurrently, the selection rate for the incorrect model gradually decreased to 0. The impact of sample size was most pronounced for BCFI and BTLI. By a size of 200, it was unlikely for BCFI and BTLI to choose the incorrect model or fail to make a choice. The impact of sample size on BRMSEA was the second highest, except under the smallest factor loading prior variance (N3). In this case, BRMSEA approached 100% accuracy for sample sizes of 300 or greater. The performance of DIC showed minimal impact from sample size, and the choice of prior had a greater influence on DIC model selection.

The impact of factor loading priors was most pronounced at a sample size of 100 when using $N(0, 0.01)$, similar to the effect observed when the incorrect model is an overfitting model. Specifically, $N(0, 0.01)$ had the greatest impact on BRMSEA, resulting in a decrease in its correct model selection rate and an increase in the probability of not making a selection. Although the proportion of BRMSEA selecting the correct model also increased with sample size, even when the sample size reached 500, about 40% of the BRMSEA under $N(0, 0.01)$ failed to make a selection. For sample sizes below 200, BCFI and BTLI were affected by $N(0, 0.01)$ and exhibited a slightly higher rate of incorrect model selection. For sample sizes from 300 to 500, BCFI and BTLI still had the probability of not choosing. However, the factor loading priors had minimal impact on DIC, whose incorrect model selection rate remains at 50%.

Regarding latent factor matrix priors, similarities were observed in performance between conditions with IW prior and SS prior. That is, the noninformative and informative priors yielded different effects. When using the noninformative prior, the rate of incorrect model selection rate for each fit index was 0, while the correct model selection rate increased with the sample size. However, when using the informative priors, the incorrect model selection rate increased, especially in small samples. The incorrect model selection rate was highest for DIC and approximately 25% for BCFI, BTLI, and BRMSEA. The influence of the informative prior decreased as the sample size increased. However, the impact on DIC for all sample size levels remained the same, and the incorrect model selection rate remained constant as the sample size increased.

Moreover, the IW prior and SS prior had different effects manifested in how Bayesian approximate fit indices respond based on the level of prior information and the impact on DIC. When the sample size was 100, the $IW(I, 25)$ (IW4) showed different results compared with the other two priors with lower degrees of freedom (IW2 and IW3). The rates of incorrect model selection for BCFI, BTLI, and BRMSEA increased and exceeded the correct model selection rates (Fig. 8A). In particular, the incorrect model selection rates for BCFI and BTLI were approximately 60%. In comparison, BRMSEA tends to have a proportion of around 40% in being unable to choose between the two models. However, under the SS prior (Fig. 8B), the IG distribution priors' different shapes and scale hyperparameters exhibit similar effects. Regarding DIC, under the IW prior (Fig. 8A), the correct model selection rates were consistently higher than the incorrect model selection rates, while under the SS prior (Fig. 8B), the incorrect model selection rates for DIC were similar to or even slightly higher than the correct model selection rates.

5.2.2.2 The Correct Model M3 with the Incorrect Model M1 When analyzing the data generated by M3 using M1 which ignored two correlations, we were interested in whether the Bayesian model fit indices are able to select the correct model (M3) rather than the incorrect one (M1). We compared the differences in model selection across different fit indices (Fig. 9).

The sample size and prior distribution had minimal impact on the Bayesian model fit indices, and the rates of correctly selecting the model were close to 1 under all conditions. Except for the SS prior in the latent factor matrix under the

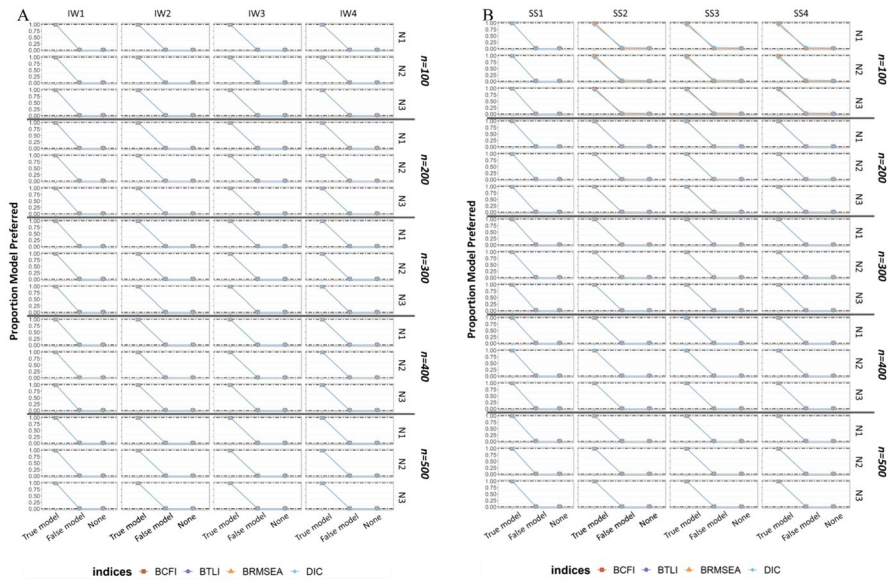


Fig. 9 Model selection rates between the correct model M3 and the incorrect model M1. *Note.* In the horizontal axis, "True model" represents selecting the correct model (M3); "False model" represents selecting the incorrect model (M1); "None" indicates that the fit indices did not choose between the correct and incorrect models. The abbreviations N1–N3 represented $N(0, \infty)$, $N(0, 10)$, and $N(0, 0.01)$; IW1–IW4 represented $IW(0, -4)$, $IW(I, 3)$, $IW(I, 9)$, and $IW(I, 25)$; and SS1–SS4 represented $IG(-1, 0)$, $IG(0.001, 0.001)$, $IG(0.01, 0.01)$, and $IG(0.1, 0.1)$. Bold black lines separated different sample sizes represented by " n "

informative prior conditions (SS2, SS3, and SS4), the correct model selection rate for DIC decreased slightly but remained close to 1. It was indicated that when the correct model was M3 and the incorrect model was M1, all Bayesian model fit indices could select the correct model across different conditions.

6 Empirical Study

6.1 Background

We utilized the classic dataset by Holzinger and Swineford (1939), which is widely applied in many studies, books, and software manuals related to structural equation modeling. The dataset is embedded within the Blavaan package in R, developed by Merkle and Rosseel (2015). The dataset includes intelligence test scores from 301 children in the seventh and eighth grades from two different schools: Pasteur School and Grant-White School.

6.2 Method

6.2.1 Materials

The intelligence test consists of 26 variables; only a subset of nine variables is included in the built-in data of the Blavaan package since these nine variables are more commonly used in the literature. These nine variables include Visual Perception, Cubes, Lozenges, Paragraph Comprehension, Sentence Completion, Word Meaning, Addition, Counting Dots, and Curved Capitals, coded as y1–y9. Higher scores indicate higher proficiency in the corresponding variable. Moreover, there are three latent factors: Spatial Ability, Verbal Ability, and Speed, coded as f1, f2, and f3. Each latent factor comprises three variables: y1–y3 measuring f1, y4–y6 measuring f2, and y7–y9 measuring f3.

We referred to Jöreskog (1969) when exploring the factor structure of this intelligence test dataset. We exclusively utilized data from 144 participants from the Grant-White School for our empirical study to avoid the influence of school variation on the results.

6.2.2 Analysis

Reliability and descriptive analysis were conducted using SPSS version 23.0. Model evaluation was performed using *Mplus* version 8.6. The fit indices employed for frequentist estimation included CFI, TLI, and RMSEA. A model is considered a good fit when the CFI and TLI are larger than 0.95 and the RMSEA is less than 0.06 (Hu and Bentler 1999). We evaluated the fit using BCFI, BTLI, BRMSEA, and PPp values as fit indices in Bayesian estimation. The model selection process involved the evaluation of fit indices such as BCFI, BTLI, BRMSEA, and DIC. The cutoff values for assessing model fit and conducting model selection remained consistent with those in the simulation studies. Furthermore, the prior distributions applied were identical to those used in the simulation studies (refer to Table 1).

Table 2 Results of descriptive and reliability analysis in the empirical study

Latent factors	M(SD)	Cronbach's α
Spatial Ability (f1)	4.374(0.854)	0.665
Verbal Ability (f2)	3.495(1.025)	0.878
Speed (f3)	4.909(0.854)	0.761
The intelligence test	4.259(0.689)	0.808

M represents the mean scores of the items in the subscale, and *SD* represents the sample standard deviation. Sample size is 144

Table 3 The modification process of model using ML estimation

Models	χ^2	df	$\Delta\chi^2$	CFI	TLI	RMSEA
The original model	51.876	24		0.941	0.911	0.09
Adding cross-loading: f1 by y9						
The modified model	28.341	23	23.535	0.989	0.982	0.04

The f1 represents the latent factor of Spatial Ability and the y9 represents the variable Curved Capitals

6.3 Results

The reliability was good, as Cronbach's α indicated (see Table 2). The data were treated as normally distributed.

6.3.1 Model Fitting Under the ML Estimation

Confirmatory factor analysis was conducted on the data using ML estimation. Subsequently, the model was iteratively modified based on the modified index (MI) until the fit indices met the criteria for a satisfactory fit. The model fit results obtained during the iterative modification process are presented in Table 3. After adding one cross-loading parameter, all the fit indices indicated a good model fit. In the subsequent Bayesian estimation, the modified model was utilized as the correctly fitted model for the given data.

6.3.2 Model Fitting Under the Bayesian Estimation

The impact of different prior distributions on Bayesian model fit indices was consistent with the findings from Simulation Study 1 (refer to Table 4). The prior distribution assigned for factor loadings had a substantial influence. Specifically, when the $N(0, 0.01)$ condition was applied, notable decreases were observed in the BCFI, BTLI, BRMSEA, and PPp -values, indicating a poor model fit of the model. BCFI and BTLI exhibited a decline of approximately 0.2 compared to the uninformative prior for factor loadings, while BRMSEA has increased by approximately 0.1, and all PPp -values reached 0.

Setting the IW prior for the latent factor matrix yielded different effects based on the degrees of freedom. In contrast, in the case of the SS prior, changing the shape and scale hyperparameters of IG distribution did not produce different effects.

When the IW prior was set as $IW(I, 25)$, all BCFI, BTLI, BRMSEA, and PPp -values substantially decreased, falling below the cutoff value. The effect of the IW prior is smaller than that of the normal prior for factor loadings. Compared to the uninformative IW prior, BCFI and BTLI experienced a decrease of approximately 0.1. Additionally, the BRMSEA increased by approximately 0.07, while the PPp -value was slightly below the predetermined cutoff value of 0.05 but did not decrease to 0. Furthermore, when the $IW(I, 9)$ prior was applied, although the Bayesian model fit

Table 4 Results of model fit evaluation in the empirical study

Prior settings		Bayesian model fit indices			
Factor loadings	Latent factor matrix	BCFI	BTLI	BRMSEA	PPp
For the entire matrix					
N1	IW1	0.987	0.981	0.042	0.336
N1	IW2	0.987	0.981	0.041	0.340
N1	IW3	0.979	0.969	0.053	0.250
N1	IW4	0.901	0.858	0.113	0.003
N2	IW1	0.987	0.981	0.042	0.338
N2	IW2	0.988	0.982	0.041	0.341
N2	IW3	0.980	0.970	0.052	0.253
N2	IW4	0.906	0.862	0.112	0.005
N3	IW1	0.773	0.722	0.159	0.000
N3	IW2	0.775	0.721	0.159	0.000
N3	IW3	0.774	0.717	0.160	0.000
N3	IW4	0.504	0.454	0.222	0.000
Using the separation strategy					
N1	SS1	0.989	0.983	0.039	0.340
N1	SS2	0.989	0.983	0.039	0.336
N1	SS3	0.989	0.983	0.039	0.336
N1	SS4	0.989	0.984	0.038	0.338
N2	SS1	0.989	0.984	0.039	0.341
N2	SS2	0.989	0.984	0.038	0.337
N2	SS3	0.989	0.984	0.038	0.336
N2	SS4	0.989	0.984	0.038	0.338
N3	SS1	0.771	0.718	0.160	0.000
N3	SS2	0.772	0.718	0.160	0.000
N3	SS3	0.772	0.718	0.160	0.000
N3	SS4	0.772	0.718	0.160	0.000

The bold indicates that the value of the Bayesian model fit index showing poor model fit

indices did not fall below the cutoff value, a substantial decrease was still observed. BCFI decreased by approximately 0.1, BTLI decreased by approximately 0.2, BRMSEA increased by approximately 0.01, and PPp value decreased by approximately 0.1.

6.3.3 Model Selection Under the Bayesian Estimation

When the incorrect model was more complex (overfitted), an additional cross-loading from f2 to y1 was added. However, when the incorrect model was misspecified, the cross-loading f1 to y9 included in the modified (correct) model was disregarded.

Table 5 Results of model selection when adding an additional cross-loading in the empirical study

Prior settings		Bayesian model fit indices			
Factor loadings	Latent factor matrix	Δ BCFI	Δ BTLI	Δ BRMSEA	Δ DIC
For the entire matrix					
N1	IW1	0.002	0.004	-0.003	-1.223
N1	IW2	0.002	0.004	-0.004	-1.509
N1	IW3	<u>0.005</u>	<u>0.007</u>	-0.006	-2.210
N1	IW4	-0.002	-0.002	0.000	0.604
N2	IW1	0.002	0.004	-0.003	-1.239
N2	IW2	0.003	0.004	-0.004	-1.502
N2	IW3	0.004	0.007	-0.006	-2.150
N2	IW4	-0.003	-0.003	0.001	0.601
N3	IW1	-0.016	-0.014	0.004	6.458
N3	IW2	-0.015	-0.012	0.003	5.814
N3	IW3	-0.015	-0.008	0.002	5.019
N3	IW4	-0.019	-0.013	0.002	9.520
Using the separation strategy					
N1	SS1	0.002	0.003	-0.004	-1.456
N1	SS2	0.003	0.004	-0.004	-1.195
N1	SS3	0.003	0.004	-0.004	-1.196
<u>N1</u>	<u>SS4</u>	0.003	<u>0.005</u>	-0.005	-1.325
N2	SS1	0.002	0.004	-0.004	-1.428
N2	SS2	0.003	0.004	-0.005	-1.161
N2	SS3	0.003	0.004	-0.005	-1.162
N2	SS4	0.002	0.004	-0.005	-1.290
N3	SS1	-0.017	-0.018	0.005	7.626
N3	SS2	-0.017	-0.017	0.005	7.361
N3	SS3	-0.017	-0.017	0.005	7.361
N3	SS4	-0.017	-0.017	0.005	7.375

The underlines indicate choosing the correct model, the bold indicates choosing the incorrect model, and the others indicate choosing none of the models

6.3.3.1 Overfitted Model The findings in this scenario were consistent with those of Simulation Study 2 (Table 5). Specifically, the influence of priors for factor loading was greater than that for the latent factor matrix. The priors for factor loadings greatly influenced BCFI and BTLI. When setting as $N(0, 0.01)$, both BCFI and BTLI selected the incorrect model. DIC was less affected. Only when IW($I, 25$) or any SS prior was set for the latent factor matrix at the same time could $N(0, 0.01)$ affect DIC to select the incorrect model. For priors with less information, BCFI, BTLI, and DIC were less affected and tended to either not select or select the correct model. BRMSEA was unable to distinguish between the two models under all prior conditions. Following the principle of parsimony, researchers should choose the more parsimonious model.

Table 6 Results of model selection when neglecting true cross-loading in the empirical study

Prior settings		Bayesian model fit indices			
Factor loadings	Latent factor matrix	Δ BCFI	Δ BTLI	Δ BRMSEA	Δ DIC
For the entire matrix					
N1	IW1	0.048	0.068	−0.047	−21.640
N1	IW2	0.048	0.066	−0.047	−20.789
N1	IW3	0.048	0.066	−0.041	−20.719
N1	IW4	0.029	0.032	−0.013	−12.350
N2	IW1	0.048	0.068	−0.047	−21.670
N2	IW2	0.049	0.067	−0.047	−20.798
N2	IW3	0.048	0.066	−0.041	−20.740
N2	IW4	0.028	0.032	−0.013	−12.551
N3	IW1	0.084	0.108	−0.028	−40.113
N3	IW2	0.083	0.106	−0.028	−39.718
N3	IW3	0.092	0.131	−0.034	−45.503
N3	IW4	0.028	0.028	−0.006	−14.750
Using the separation strategy					
N1	SS1	0.050	0.070	−0.050	−22.295
N1	SS2	0.050	0.069	−0.050	−22.205
N1	SS3	0.050	0.069	−0.050	−22.199
N1	SS4	0.050	0.070	−0.051	−22.193
N2	SS1	0.050	0.071	−0.049	−22.279
N2	SS2	0.050	0.070	−0.050	−22.177
N2	SS3	0.050	0.070	−0.050	−22.173
N2	SS4	0.050	0.070	−0.050	−22.175
N3	SS1	0.080	0.089	−0.023	−37.025
N3	SS2	0.080	0.089	−0.023	−37.007
N3	SS3	0.080	0.089	−0.023	−37.007
N3	SS4	0.080	0.089	−0.023	−37.002

The underlines indicate choosing the correct model, and the others indicate choosing none of the models

6.3.3.2 Misspecified Model Generally, the proportion of Bayesian model fit indices selecting the correct model was much higher than in the previous scenario (Table 6). BCFI, BTLI, and DIC consistently selected the correct model across all prior conditions. However, when the factor loading prior was set as $N(0, 0.01)$ and the latent factor matrix prior was $IW(I, 25)$, BRMSEA faced difficulty selecting the correct and incorrect models. Similar to Simulation Study 2, BRMSEA was influenced by priors to tend indecisiveness. In this particular scenario, the incorrect model was more parsimonious than the correct one. Following the principle of parsimony, BRMSEA should select the more parsimonious but incorrect model, leading to a wrong model selection.

7 Discussion

7.1 Impacts of Prior on Evaluating Model Fit

Study 1 utilized Monte Carlo simulations to evaluate the performance of different Bayesian model fit indices (BCFI, BTLI, BRMSEA, and PPp value) in assessing model fit across different priors for factor loadings and the variance–covariance matrix of latent factors at different sample sizes. As the sample size increased, all fit indices demonstrated good model fit. Specifically, BCFI and BTLI were the most influenced by the sample size, followed by BRMSEA, while the PPp -values were the least affected. These findings were consistent with previous studies (Liang 2020; Liu et al. 2022; Winter and Depaoli 2022a 2022b). Following the recommendations of Asparouhov and Muthén (2021), it was more appropriate to use Bayesian approximate fit indices and PPp -values to assess model fit when the sample size was larger than 200. Bayesian model fit indices were more susceptible to prior distributions in smaller sample sizes, potentially resulting in a poor fit for the correct model and leading to the incorrect rejection of the true model in practical applications.

When extreme errors were present in the priors of factor loadings and the variance–covariance matrix of latent factors, such as normal priors with small variance and IW priors with large df , the Bayesian model fit indices were severely affected, indicating poor model fit. This result was expected, and previous studies also found that inaccurate informative prior would lead to bias in model parameter estimation (Depaoli 2014; McNeish 2016; Winter and Depaoli 2022c). The priors used in our study were all inaccurate priors with a mean of 0, and small variance substantially amplified such inaccuracies. Therefore, when researchers are unsure about the information in the priors, it is more conservative to specify a larger prior variance for the parameters (Depaoli 2014; Zheng et al. 2022).

The priors had the greatest impact on PPp -values, followed by BRMSEA, and had the most negligible impact on BCFI and BTLI. However, the impact of the priors for factor loading was more pronounced. For PPp -values, using $N(0, 0.01)$ led to a poor model fit across all sample size levels. For BRMSEA, $N(0, 0.01)$ resulted in a poor model fit even when the sample size reached 400. BCFI and BTLI were less likely to indicate poor model fit under different priors in larger sample sizes. This finding was consistent with Winter and Depaoli (2022a), which investigated the effects of prior specification on BCFI, BTLI, BRMSEA, and PPp -values in the latent growth models. It was shown that the performance of Bayesian model fit indices was relatively stable when assessing model fit across different priors and model frameworks.

Additionally, for the variance–covariance matrix of latent factors, different shape and scale hyperparameters of the SS prior had a relatively consistent impact on each fit index. van Erp et al. (2018) found that $IG(0.001, 0.001)$, $IG(0.01, 0.01)$, and $IG(0.1, 0.1)$ did have a substantial influence on both the parameter estimation and convergence rate of BSEM. However, in Simulation Study 1, when $IG(0.001, 0.001)$, $IG(0.01, 0.01)$, and $IG(0.1, 0.1)$ were used for the diagonal

elements of the latent factor matrix, there was no discernible difference in the performance of model fit evaluation. Hence, the beliefs reflected by the selected hyperparameters may be relatively ambiguous for the prior and are unlikely to lead to the rejection of the correct model by model fit indices. Moreover, there have not been specific rules in the SS prior for the shape and scale hyperparameters of the diagonal elements' IG distribution (van Erp et al. 2018). Hence, the information level in the inverse Gamma distribution prior is less intuitive than in the normal priors. In the context of future research, it is meaningful to offer precise elucidation and specifications for the IG prior and to explore the impact of IG prior with various hyperparameters on model fitting indices.

7.2 Impacts of Prior on Model Selection

Study 2 included two additional data generation and fitting models compared with Study 1. As the sample size increased, the likelihood of all fit indices being unable to distinguish between the correct and incorrect models increased, mainly when the incorrect model was more complex. When the incorrect model was more parsimonious, the likelihood of fit indices accurately selecting the correct model increased as the sample size increased. With extreme errors in the priors for factor loadings and latent factor variance–covariance matrix, incorrect model selection rates increased, especially in small sample sizes. Although the impact of the prior distributions on the latent factor matrix was not evident in the assessment of model fit, informative priors had a substantially different impact on the results compared to uninformative priors in model selection.

7.2.1 Overfitted Model

When the incorrect model was an overfitting complex model with two additional cross-loadings, all priors for the parameters had the greatest impact on BCFI and BTLI in small sample sizes. In contrast, DIC had a slightly smaller impact, and BRMSEA was the least affected. However, BCFI, BTLI, and DIC may select the incorrect model only under an extreme deviation before factor loadings. In particular, when applying an IW prior (instead of an SS prior) for the variance–covariance matrix of latent factors, the $N(0, 0.01)$ prior suddenly increased the accuracy of DIC in selecting the correct model when the sample size was more extensive than or equal to 400, showing a tendency to select more parsimonious models, consistent with previous research findings (Winter and Depaoli 2022b). However, Liu et al. (2022) pointed out that DIC preferred complex models when the incorrect model was more complex in model selection, which contradicted the existing results. Liu et al. (2022) also found that setting $N(0, 0.01)$ prior to the factor loading would cause abnormal pattern changes in DIC: As the sample size increased, the size of Δ DIC between $N(0, 0.01)$ and $N(0, 1000)$ did not change linearly. It may indicate that both sample sizes and priors influence DIC. The phenomenon has not yet been able to be summarized regularly. Further investigation is required to understand the mechanisms and patterns underlying this influence comprehensively.

When the overfitted model included two additional residual correlations, the influence of sample size and prior conditions became more apparent. The prior distributions for factor loadings had the greatest impact on BRMSEA, followed by BCFI and BTLI, with almost no impact on DIC. The IW and SS priors for the variance–covariance matrix of latent factors had the greatest impact on DIC, followed by BRMSEA, with a minor impact on BCFI and BTLI. DIC maintained a selection rate of around 50% for correct and incorrect models, regardless of sample size. BCFI, BTLI, and BRMSEA exhibited lower probabilities for small sample sizes than DIC in selecting correct or incorrect models. Moreover, they were more likely unable to select between two models as the sample size increased. Our findings are consistent with Winter and Depaoli (2022b).

Our findings indicated that Bayesian model fit indices, particularly Bayesian approximate fit indices, often fail to differentiate between overfitting models and the correct model across most conditions. According to the principle of parsimony, researchers should actively choose the more parsimonious model when two models exhibit similar fit, as it is likely to be the correct model in cases where the alternative model is overfitting. DIC had the highest sensitivity for model selection and could differentiate between models with additional residual correlations incorporating informative priors. However, this came with the cost of increased false positive rates, indicating low reliability in using DIC for model selection. These findings are consistent with previous studies on Bayesian model selection indices (Lu et al. 2017).

7.2.2 Misspecified Model

When the incorrect model was misspecified, all Bayesian model fit indices showed a stronger tendency to select the correct model, even in small samples. Our findings are consistent with previous research findings (Asparouhov and Muthén 2021; Cain and Zhang 2019; Winter and Depaoli 2022a, 2022b).

When the incorrect model failed to account for two cross-loadings under the IW prior (instead of the SS prior), Bayesian approximate fit indices exhibited sensitivity to the df of informative priors. This pattern closely mirrored the findings of Simulation Study 1, indicating the selecting information level of the SS prior was insufficient to manifest in model selection performance. BRMSEA was most affected by the priors for factor loadings and failed to select between the two models, even in large samples. However, in this situation, according to the principle of parsimony, the incorrect model that ignored cross-loadings was more parsimonious. Therefore, when BRMSEA cannot determine which model fits better, it may lead to an erroneous choice. It is recommended that researchers should choose a model with meaningful implications based on the specific context rather than solely relying on the principle of parsimony (Cain and Zhang 2019). DIC was notably influenced by the latent factor matrix priors, which had the highest selection rate for the incorrect model among all the fit indices. The result was consistent with Depaoli et al. (2023) but contradicted the view that DIC is an information indicator suitable for Bayesian model selection (see, e.g., Asparouhov et al. 2015; Liang and Luo 2020; Pokropek 2020). Consequently, using DIC for model

selection under prior conditions could be unreliable. Conversely, both BCFI and BTLI can select the correct model reasonably well.

Under all conditions, when the incorrect model neglected two residual correlations, BCFI, BTLI, BRMSEA, and DIC consistently had selection rates of approximately 100% for the correct model. The result suggested that neglecting residual correlations represents a more severe error than neglecting cross-loadings. In this scenario, the impact of priors is minimal, and the model selection performance of Bayesian model fit indices is stable, with very high accuracy.

8 Insights and Suggestions

Firstly, when the sample size is less than 200, extreme priors may notably reduce the performance of Bayesian model fit indices in assessing model fit and selecting models, thereby leading to biased results. Consequently, it is recommended that researchers conduct sensitivity analyses to understand the impact of priors and compare the results of various model fit indices and model selection criteria to make well-informed decisions.

Depaoli and van de Schoot (2017) suggest that researchers should report the results of sensitivity analyses transparently and clearly. If the Bayesian model fit index fluctuates greatly in the sensitivity analysis, we cannot change the original priors based on this result, which is an outcome-oriented operation. Researchers should report and discuss prior fluctuations to enhance subsequent researchers' understanding of current outcomes. If it is suspected that the original model setting or prior needs to be changed due to a model setting error or prior selection error, an expert should be consulted, and the study should be re-conducted according to the WAMBS-Checklist (Depaoli and van de Schoot 2017).

Secondly, for model fit evaluation, although Bayesian approximate fit indices can also be affected by sample sizes and prior conditions, they generally provide more accurate assessments of model fit compared to *PPp*-values. Among them, BCFI and BTLI may be the most suitable choices for model evaluation.

Lastly, regarding model selection, DIC is greatly influenced by priors. Although DIC does not encounter situations where it cannot make a selection, it has similar rates of correct and incorrect selections, potentially leading to biased model selection results. Bayesian approximate fit indices are less influenced by priors and have lower rates of incorrect selections. Consequently, Bayesian approximate fit indices are more accurate model selection criteria than DIC. However, a limitation arises when they sometimes struggle to select the correct and incorrect models, especially in the case of BRMSEA. In such situations, researchers should exercise caution when relying solely on the principle of parsimony and instead consider the substantive meaning of the model when making selection decisions.

9 Further Research

The current study considered a situation where the data were continuous and followed a normal distribution. Previous research results have shown the impact of the data type on parameter estimation in BSEM, particularly concerning categorical data (Lee et al. 2020). Future research can investigate the impact of priors on Bayesian model fit indices with different types of data.

For prior settings, we selected default noninformative priors in *Mplus* and priors deviated from true value in the present study. In future research, it would be valuable to incorporate correct informative priors consistent with the population model as new prior conditions. Additionally, the SS prior can be set using various methods (Liu et al. 2022), and further exploration is needed to determine whether using different SS priors from others would yield different effects compared to overall priors.

In addition, we set the same priors for cross-loadings and primary factor loadings in Simulation Study 2. Future research may explore whether priors will still substantially impact Bayesian model fit indices when only cross-loading priors are incorrect and main factor loading priors are correct. Alternatively, it can be reversed and explored when the primary factor loading priors are incorrect, and the cross-loading priors are correct. More simulation studies could build on the current research to explore these questions further.

For fit indices, approximate fit indices were not intended for use as model selection criteria, and there is a lack of consistent standards for determining cutoff values (Cheung and Rensvold 2002; Winter and Depaoli 2022b; Marshall et al. 2022). Moreover, the choice of cutoff values might need to be adjusted based on sample size (Chen 2007). In the present study, Bayesian approximate fit indices were more likely to encounter difficulty making selections compared to DIC. This phenomenon may be due to sensitivities to priors and inaccurate cutoff values. Future research should explore whether there are fixed cutoff values suitable for model selection using Bayesian approximate fit indices. Moreover, it is worth investigating the dynamic cutoff values for Bayesian model fit indices (McNeish and Wolf 2023).

Appendix

Bayesian approximate fit indices are relatively new model fit indices under Bayesian framework compared to the commonly used *PPp* value and DIC, so we hope to provide readers with formulas to help them understand how these Bayesian approximate fit indices are calculated.

BRMSEA was the first Bayesian approximate fitting index corresponding to the frequency framework RMSEA proposed by Hoofs et al. (2018). The formula can be written as follows:

$$\text{BRMSEA}_s = \sqrt{\max\left(0, \frac{D_{H,s}^{\text{obs}} - p^*}{(p^* - pD_0)N}\right)}, \quad (3)$$

where s is the s -th iteration in MCMC estimation. $D_{H,s}^{\text{obs}}$ is the hypothesized model discrepancy function calculated for observation data in the s -th MCMC iteration. $df = p^* - q$, where p^* is the number of nonredundant sample moment and q represents the number of estimated parameters. In models with mean and covariance structures, $p^* = \frac{p(p+3)}{2}$, where p is the number of variables in the model. pD_0 is a penalty term for model complexity, reflecting the number of valid parameters in the null model.

BCFI and BTLI are Bayesian approximate fitting indices further inferred by Garnier-Villarre and Jorgensen (2020) on the basis of BRMSEA, and their calculation processes are similar. For BCFI,

$$\text{BCFI}_s = 1 - \frac{(D_{H,s}^{\text{obs}} - pD_H) - (p^* - pD_H)}{(D_{0,s}^{\text{obs}} - pD_0) - (p^* - pD_0)} = 1 - \frac{D_{H,s}^{\text{obs}} - p^*}{D_{0,s}^{\text{obs}} - p^*}, \quad (4)$$

where $D_{0,s}^{\text{obs}}$ is the baseline model discrepancy function. pD_H reflects the number of valid parameters in the null model.

For BTLI,

$$\text{BTLI}_s = \frac{\frac{D_{0,s}^{\text{obs}} - pD_0}{p^* - pD_0} - \frac{D_{H,s}^{\text{obs}} - pD_H}{p^* - pD_H}}{\frac{D_{0,s}^{\text{obs}} - pD_0}{p^* - pD_0} - 1}. \quad (5)$$

The posterior distribution of BRMSEA, BCFI, and BTLI will be obtained from the BRMSEA_s , BCFI_s , and BTLI_s computed over multiple iterations of MCMC. The median of the posterior distribution is used as a point estimate for BRMSEA, BCFI, and BTLI. The 90% confidence interval can also be obtained from the posterior distribution.

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Code availability The code used for analysis in this study is available.

Data availability The data that support the findings of this study are available.

Declarations

Conflict of interest The authors have no conflicts of interest.

References

- Asparouhov, T., and Muthén, B. 2010. Bayesian analysis of latent variable models using Mplus.
- Asparouhov, T., and B. Muthén. 2021. Advances in Bayesian Model Fit Evaluation for Structural Equation Models. *Structural Equation Modeling: A Multidisciplinary Journal* 28 (1): 1–14. <https://doi.org/10.1080/10705511.2020.1764360>.
- Asparouhov, T., B. Muthén, and A.J.S. Morin. 2015. Bayesian Structural Equation Modeling With Cross-Loadings and Residual Covariances: Comments on Stromeier et al. *Journal of Management* 41 (6): 1561–1577. <https://doi.org/10.1177/0149206315591075>.
- Baldwin, S.A., and G.W. Fellingham. 2013. Bayesian Methods for the Analysis of Small Sample Multi-level Data with a Complex Variance Structure. *Psychological Methods* 18 (2): 151–164. <https://doi.org/10.1037/a0030642>.
- Cain, M.K., and Z. Zhang. 2019. Fit for a Bayesian: An Evaluation of PPP and DIC for Structural Equation Modeling. *Structural Equation Modeling: A Multidisciplinary Journal* 26 (1): 39–50. <https://doi.org/10.1080/10705511.2018.1490648>.
- Chen, F.F. 2007. Sensitivity of Goodness of Fit Indexes to Lack of Measurement Invariance. *Structural Equation Modeling: A Multidisciplinary Journal* 14 (3): 464–504. <https://doi.org/10.1080/10705510701301834>.
- Cheung, G.W., and R.B. Rensvold. 2002. Evaluating Goodness-of-fit Indexes for Testing Measurement Invariance. *Structural Equation Modeling* 9 (2): 233–255. https://doi.org/10.1207/S15328007SEM0902_5.
- Depaoli, S. 2014. The Impact of Inaccurate “Informative” Priors for Growth Parameters in Bayesian Growth Mixture Modeling. *Structural Equation Modeling: A Multidisciplinary Journal* 21: 239–252.
- Depaoli, S. 2021. *Bayesian structural equation modeling*. The Guilford Press.
- Depaoli, S., and J.P. Clifton. 2015. A Bayesian Approach to Multilevel Structural Equation Modeling With Continuous and Dichotomous Outcomes. *Structural Equation Modeling: A Multidisciplinary Journal* 22 (3): 327–351. <https://doi.org/10.1080/10705511.2014.937849>.
- Depaoli, S., and R. van de Schoot. 2017. Improving transparency and replication in Bayesian statistics: The WAMBS-Checklist. *Psychological Methods* 22 (2): 240–261. <https://doi.org/10.1037/met0000065>.
- Depaoli, S., H. Liu, and L. Marvin. 2021. Parameter Specification in Bayesian CFA: An Exploration of Multivariate and Separation Strategy Priors. *Structural Equation Modeling: A Multidisciplinary Journal* 28 (5): 699–715. <https://doi.org/10.1080/10705511.2021.1894154>.
- Depaoli, S., S.D. Winter, and H. Liu. 2023. Under-Fitting and Over-Fitting: The Performance of Bayesian Model Selection and Fit Indices in SEM. *Structural Equation Modeling: A Multidisciplinary Journal*. <https://doi.org/10.1080/10705511.2023.2280952>.
- Edwards, K.D., and T.R. Konold. 2022. Impact of Informative Priors on Model Fit Indices in Bayesian Confirmatory Factor Analysis. *Structural Equation Modeling: A Multidisciplinary Journal* 30: 272–283. <https://doi.org/10.1080/10705511.2022.2126359>.
- Garnier-Villareal, M., and T.D. Jorgensen. 2020. Adapting fit Indices for Bayesian Structural Equation Modeling: Comparison to Maximum Likelihood. *Psychological Methods* 25 (1): 46–70. <https://doi.org/10.1037/met0000224>.
- Guo, J., H.W. Marsh, P.D. Parker, T. Dicke, O. Lüdtke, and T.M.O. Diallo. 2019. A Systematic Evaluation and Comparison Between Exploratory Structural Equation Modeling and Bayesian Structural Equation Modeling. *Structural Equation Modeling: A Multidisciplinary Journal* 26 (4): 529–556. <https://doi.org/10.1080/10705511.2018.1554999>.
- Hallquist, M.N., and J.F. Wiley. 2018. *MplusAutomation*: An R Package for Facilitating Large-Scale Latent Variable Analyses in M plus. *Structural Equation Modeling: A Multidisciplinary Journal* 25 (4): 621–638. <https://doi.org/10.1080/10705511.2017.1402334>.
- Holzing, K. J., and Swineford, F. 1939. A study in factor analysis: The stability of a bi-factor solution. *Supplementary educational monographs*.
- Hoofs, H., R. Van De Schoot, N.W.H. Jansen, and Ij. Kant. 2018. Evaluating Model Fit in Bayesian Confirmatory Factor Analysis With Large Samples: Simulation Study Introducing the BRMSEA. *Educational and Psychological Measurement* 78 (4): 537–568. <https://doi.org/10.1177/0013164417709314>.

- Hu, L., and P.M. Bentler. 1999. Cutoff Criteria for fit Indexes in Covariance Structure Analysis: Conventional Criteria Versus New Alternatives. *Structural Equation Modeling: A Multidisciplinary Journal* 6 (1): 1–55. <https://doi.org/10.1080/10705519909540118>.
- Joreskog, K.G. 1969. A General Approach to Confirmatory Maximum Likelihood Factor Analysis. *Psychometrika* 34 (2, Pt. 11): 183–202. <https://doi.org/10.1007/BF02289343>.
- Kyriazos, T.A. 2018. Applied Psychometrics: Sample Size and Sample Power Considerations in Factor Analysis (EFA, CFA) and SEM in General. *Psychology* 9 (08): 2207–2230. <https://doi.org/10.4236/psych.2018.98126>.
- Lee, J., K. Jung, and J. Park. 2020. Detecting Conditional Dependence Using Flexible Bayesian Latent Class Analysis. *Frontiers in Psychology* 11: 1987. <https://doi.org/10.3389/fpsyg.2020.01987>.
- Liang, X. 2020. Prior Sensitivity in Bayesian Structural Equation Modeling for Sparse Factor Loading Structures. *Educational and Psychological Measurement* 80 (6): 1025–1058. <https://doi.org/10.1177/0013164420906449>.
- Liang, X., and Y. Luo. 2020. A Comprehensive Comparison of Model Selection Methods for Testing Factorial Invariance. *Structural Equation Modeling: A Multidisciplinary Journal* 27 (3): 380–395. <https://doi.org/10.1080/10705511.2019.1649983>.
- Liu, H., S. Depaoli, and L. Marvin. 2022. Understanding the Deviance Information Criterion for SEM: Cautions in Prior Specification. *Structural Equation Modeling: A Multidisciplinary Journal* 29 (2): 278–294. <https://doi.org/10.1080/10705511.2021.1994407>.
- Liu, H., Z. Zhang, and K.J. Grimm. 2016. Comparison of Inverse Wishart and Separation-Strategy Priors for Bayesian Estimation of Covariance Parameter Matrix in Growth Curve Analysis. *Structural Equation Modeling: A Multidisciplinary Journal* 23 (3): 354–367. <https://doi.org/10.1080/10705511.2015.1057285>.
- Lu, Z.-H., S.-M. Chow, and E. Loken. 2017. A Comparison of Bayesian and Frequentist Model Selection Methods for Factor Analysis Models. *Psychological Methods* 22 (2): 361–381. <https://doi.org/10.1037/met0000145>.
- Marsall, M., G. Engelmann, E.M. Skoda, M. Teufel, and A. Bäuerle. 2022. Measuring Electronic Health Literacy: Development, Validation, and Test of Measurement Invariance of a Revised German Version of the eHealth Literacy Scale. *Journal of Medical Internet Research* 24 (2): e28252. <https://doi.org/10.2196/28252>.
- McNeish, D. 2016. On Using Bayesian Methods to Address Small Sample Problems. *Structural Equation Modeling: A Multidisciplinary Journal* 23 (5): 750–773. <https://doi.org/10.1080/10705511.2016.1186549>.
- McNeish, D., and M.G. Wolf. 2023. Dynamic fit Index Cutoffs for Confirmatory Factor Analysis Models. *Psychological Methods* 28 (1): 61–88. <https://doi.org/10.1037/met0000425>.
- Meng, X.-L. 1994. Posterior Predictive p-Values. *The Annals of Statistics*. <https://doi.org/10.1214/aos/1176325622>.
- Merkle, E. C., and Rosseel, Y. 2015. blavaan: Bayesian structural equation models via parameter expansion. *arXiv preprint arXiv:1511.05604*.
- Muthén, B., and T. Asparouhov. 2012. Bayesian Structural Equation Modeling: A More Flexible Representation of Substantive Theory. *Psychological Methods* 17 (3): 313–335. <https://doi.org/10.1037/a0026802>.
- Pokropek, A., P. Schmidt, and E. Davidov. 2020. Choosing Priors in Bayesian Measurement Invariance Modeling: A Monte Carlo Simulation Study. *Structural Equation Modeling: A Multidisciplinary Journal* 27 (5): 750–764. <https://doi.org/10.1080/10705511.2019.1703708>.
- Schreiber, J.B., A. Nora, F.K. Stage, E.A. Barlow, and J. King. 2006. Reporting Structural Equation Modeling and Confirmatory Factor Analysis Results: A Review. *The Journal of Educational Research* 99 (6): 323–338. <https://doi.org/10.3200/JOER.99.6.323-338>.
- Schwarz, G. 1978. Estimating the Dimension of a Model. *The Annals of Statistics*. <https://doi.org/10.1214/aos/1176344136>.
- Smid, S.C., S. Depaoli, and R. Van De Schoot. 2020. Predicting a Distal Outcome Variable From a Latent Growth Model: ML versus Bayesian Estimation. *Structural Equation Modeling: A Multidisciplinary Journal* 27 (2): 169–191. <https://doi.org/10.1080/10705511.2019.1604140>.
- Sokolov, B. 2019. Sensitivity of goodness of fit indices to lack of measurement invariance with categorical indicators and many groups. *Higher School of Economics Research Paper No. WP BRP*, 86.
- Spiegelhalter, D.J., N.G. Best, B.P. Carlin, and A. Van Der Linde. 2002. Bayesian Measures of Model Complexity and Fit. *Journal of the Royal Statistical Society Series b: Statistical Methodology* 64 (4): 583–639. <https://doi.org/10.1111/1467-9868.00353>.

- van Erp, S., J. Mulder, and D.L. Oberski. 2018. Prior Sensitivity Analysis in Default Bayesian Structural Equation Modeling. *Psychological Methods* 23 (2): 363–388. <https://doi.org/10.1037/met0000162>.
- Winter, S.D., and S. Depaoli. 2022a. Performance of Model Fit and Selection Indices for Bayesian Structural Equation Modeling with Missing Data. *Structural Equation Modeling: A Multidisciplinary Journal* 29 (4): 531–549. <https://doi.org/10.1080/10705511.2021.2018656>.
- Winter, S.D., and S. Depaoli. 2022b. Sensitivity of Bayesian Model Fit Indices to the Prior Specification of Latent Growth Models. *Structural Equation Modeling: A Multidisciplinary Journal* 29 (5): 667–686. <https://doi.org/10.1080/10705511.2022.2032078>.
- Winter, S.D., and S. Depaoli. 2022. Detecting Prior-Data Disagreement in Bayesian Structural Equation Modeling. *Structural Equation Modeling: A Multidisciplinary Journal* 29 (6): 821–838. <https://doi.org/10.1080/10705511.2022.2039066>.
- Xiao, Y., H. Liu, and K.-T. Hau. 2019. A Comparison of CFA, ESEM, and BSEM in Test Structure Analysis. *Structural Equation Modeling: A Multidisciplinary Journal* 26 (5): 665–677. <https://doi.org/10.1080/10705511.2018.1562928>.
- Zheng, S., L. Zhang, Z. Jiang, and J. Pan. 2022. The Influence of Using Inaccurate Priors on Bayesian Multilevel Estimation. *Structural Equation Modeling: A Multidisciplinary Journal* 30: 429–448.
- Zondervan-Zwijnenburg, M., M. Peeters, S. Depaoli, and R. Van De Schoot. 2017. Where Do Priors Come From? Applying Guidelines to Construct Informative Priors in Small Sample Research. *Research in Human Development* 14 (4): 305–320. <https://doi.org/10.1080/15427609.2017.1370966>.

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Yonglin Feng is currently a master's student in the Department of Psychology at Sun Yat-sen University, where she also completed her undergraduate studies. Her research focuses on psychometrics, with a particular interest in the application of Bayesian methods in structural equation modeling. Yonglin has presented oral reports at several prestigious conferences, including the 2023 Meeting of the International Society for Data Science and Analytics (ISDSA) and The International Meeting of the Psychometric Society (IMPS) 2023. She has been awarded the first-class scholarship at her university and received an award for an outstanding undergraduate thesis.

Junhao Pan is a professor in the Department of Psychology at Sun Yat-sen University. He obtained his Ph.D. in Statistics from the Chinese University of Hong Kong. His primary research focuses on improving and developing statistical analysis methods for latent variable models. In recent years, he has shown particular interest in Bayesian structural equation models, as well as the integration of Bayesian regularization methods and latent variable models. His series of papers have been published in *Psychological Methods*, *Psychometrika*, *Structural Equation Modeling: A Multidisciplinary Journal* and so on.